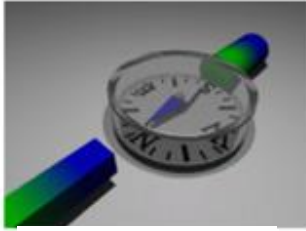


TOPIC 7.

MAGNETIC PROPERTIES

- Introduction
- Basic concepts
- Types of magnetism
- Magnetic domains
- Hysteresis cycles
- Superconductivity and magnetism.

MAGNETIC MATERIALS USES



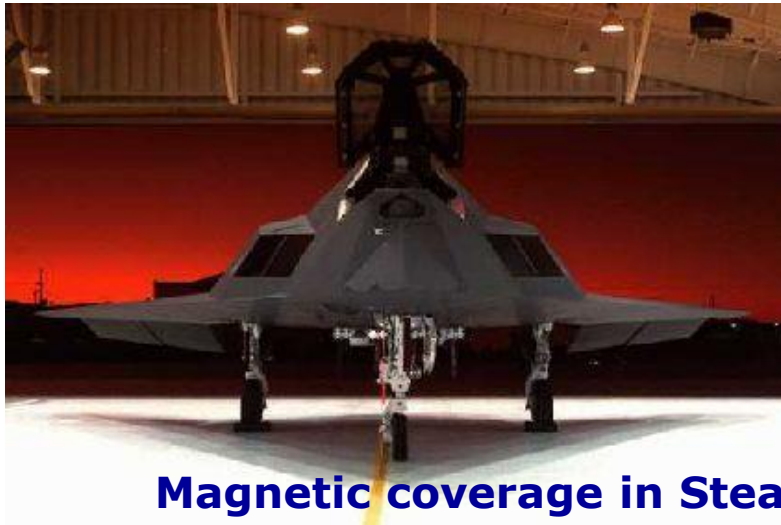
compass



Magnets



Ferrofluids



Magnetic coverage in Stealth Technologies

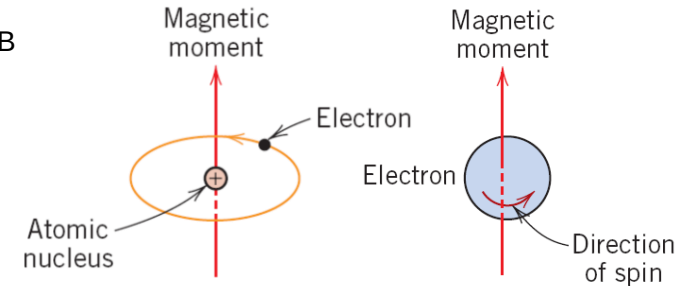
ORIGIN OF THE MAGNETIC BEHAVIOR OF MATERIALS

Magnetism = Phenomenon by which materials induce forces of attraction or repulsion over other materials.
The phenomenon has its origin on the *movement of electrically charged particles*.

The macroscopic magnetic properties of materials are a consequence of the magnetic moments associated with individual e^-

Orbital magnetic moment = $m_l \mu_B$

Spin magnetic moment = $\pm \mu_B$



Orbital

Spin

μ_B = Bohr magneton = $9,27 \cdot 10^{-24} \text{ A} \cdot \text{m}^2$
 m_l = magnetic quantum moment of e^-

Magnetic moment of an atom =

Sum of the magnetic moment of all its e^-

If the net magnetic moment of an atom is **not** null, the atom behaves like a magnetic dipole

Atomic number	Element	Electronic structure of 3d	Moment (μ_B)
21	Sc	$\uparrow$ 	1
22	Ti	$\uparrow$ $\uparrow$ 	2
23	V	$\uparrow$ $\uparrow$ $\uparrow$ 	3
24	Cr	$\uparrow$ $\uparrow$ $\uparrow$ $\uparrow$ $\uparrow$	5
25	Mn	$\uparrow$ $\uparrow$ $\uparrow$ $\uparrow$ $\uparrow$	5
26	Fe	$\uparrow\downarrow$ $\uparrow$ $\uparrow$ $\uparrow$ $\uparrow$	4
27	Co	$\uparrow\downarrow$ $\uparrow\downarrow$ $\uparrow$ $\uparrow$ $\uparrow$	3
28	Ni	$\uparrow\downarrow$ $\uparrow\downarrow$ $\uparrow\downarrow$ $\uparrow$ $\uparrow$	2
29	Cu	$\uparrow\downarrow$ $\uparrow\downarrow$ $\uparrow\downarrow$ $\uparrow\downarrow$ $\uparrow\downarrow$	0

$\uparrow$ = electronic spin orientation

BASIC CONCEPTS

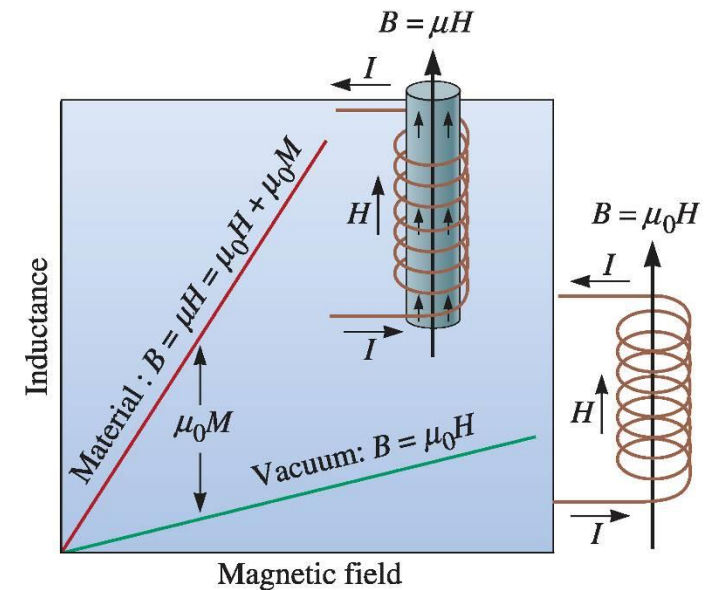
- H = **magnetic field strength** (A/m)
- B = **magnetic induction** or magnetic flux density or inductance (Wb/m² or tesla)
- μ = **permeability** of the medium through which H passes and in which B is measured (Wb/A·m or H/m)

$$B = \mu H \quad \text{and in vacuum} \quad B = \mu_0 H$$

$$\mu_0 = \text{permeability of vacuum} \quad (1.257 \times 10^{-6} \text{ H/m})$$

- μ_r = relative permeability $\mu_r = \frac{\mu}{\mu_0} \quad B = \mu H = \mu_0 H + \mu_0 M$

- M = **magnetization** (A/m) (measure of the quantity of magnetic moment per unit volume)
- χ_m = **magnetic susceptibility** $M = \chi_m H$



TYPES OF MAGNETISM

Ferromagnetic

Ferrimagnetic

Paramagnetic

$$\chi \approx 10^{-5}$$

Diamagnetic

$$\chi \approx 10^{-5}$$

Anti-ferromagnetic

non magnetizable

$$\mu_r \gg 1$$

$$\chi \approx 10^5 - 10^6$$

μ_r close to 1

Magnetic induction
(B--tesla)

$$B \cong \mu_0 M$$

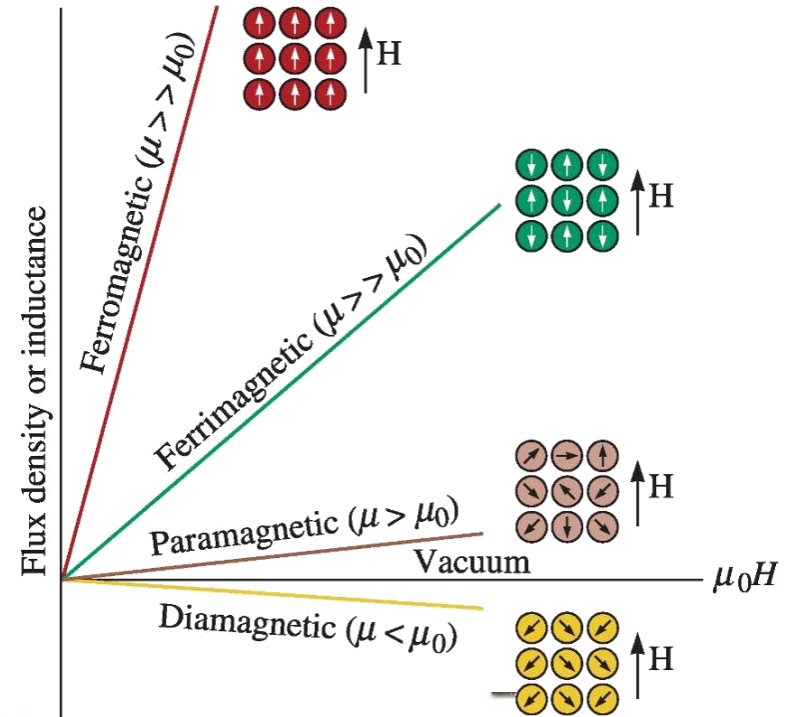
$$\hat{H} \ll M$$

(3) ferromagnetic, ferrimagnetic (χ as large as 10^6 !)
Fe₃O₄, NiFe₂O₄, ferrite(α), Co, Ni, Gd

(2) Paramagnetic
e.g., Al, Cr, Mo, Na, Ti, Zr
vacuum ($\chi = 0$)

(1) diamagnetic ($\chi \sim -10^{-5}$)
e.g., Al₂O₃, Cu, Au, Si, Ag, Zn

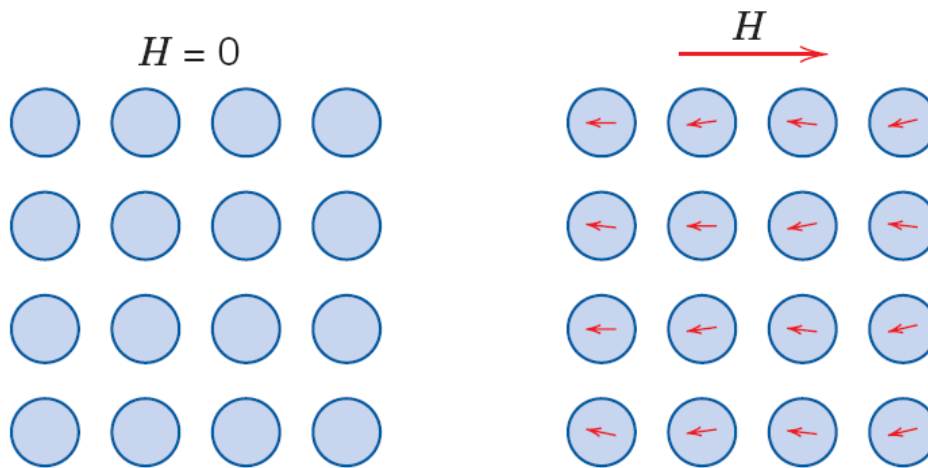
Strength of applied magnetic field (H)
(ampere-turns/m)



DIAMAGNETISM

- In the absence of an external field, no dipoles exist.
- In the presence of a field, dipoles are induced that are aligned opposite to the field direction.

(non permanent magnetism)



H = magnetic field strength

The magnetic moment induced is very small and opposite to the applied field



$$\mu_r < 1 \quad \text{and} \quad \chi_m < 0$$

$$B \text{ (diamagnetic)} \cong \mu_0 \cdot H$$

$$B \text{ (diamagnetic)} < B_0$$

The external magnetic field induces a change in the speed of the orbital movement of the e^-

Gold, silver, copper

THE FROG INCIDENT

https://www.youtube.com/watch?v=A1vyB-O5i6E&ab_channel=GrahamRumplestiltskinMcLeod

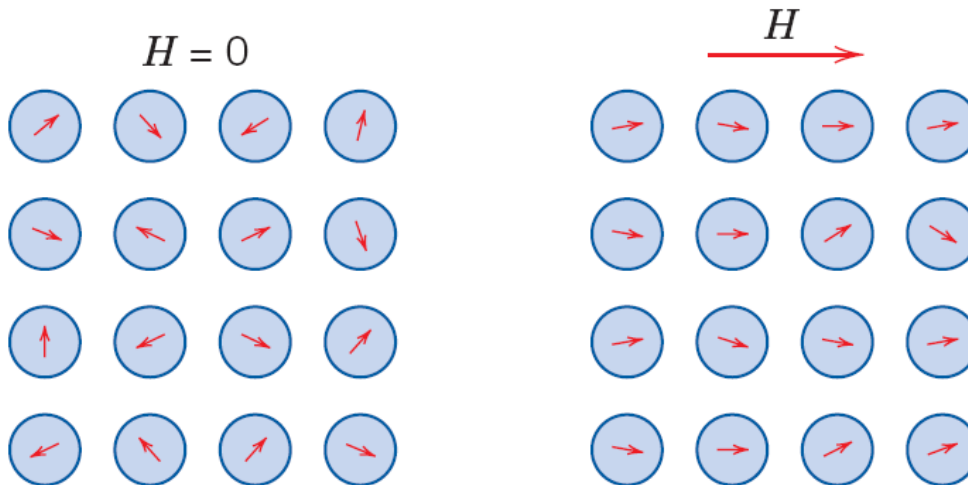


PARAMAGNETIC MATERIALS

Materials whose atoms have permanent magnetic dipoles
randomly oriented

The material does not have permanent net magnetization

When an *external field* is applied, the dipoles turn
aligning with the direction of the field



The induced magnetic field is relatively small and its effect is added to the applied field



$$\mu_r > 1 \quad \text{and} \quad 10^{-5} \leq \chi_m < 10^{-2}$$

$$B \text{ (paramagnetic)} \cong \mu_0 \cdot H$$

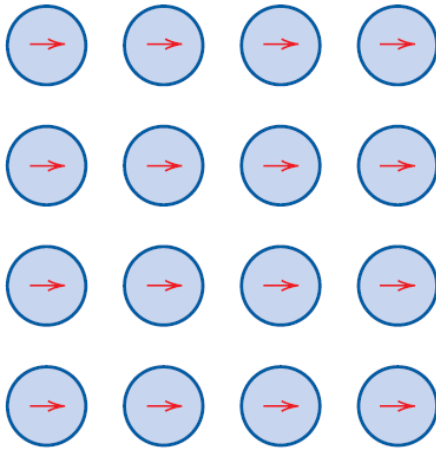
$$B \text{ (paramagnetic)} > B_0$$

Aluminium, Titanium

FERROMAGNETISM

Materials with magnetic dipoles permanently oriented in the absence of an external field. They manifest large permanent magnetization

$$H = 0$$



Alignment of atomic dipoles for a ferromagnetic material, which will exist even in the absence of an external magnetic field.

Magnetic Moment, μ_m , at 0-K for Ferromagnetic Metals.

Metal	μ_m
Fe	$2.22 \mu_B$
Co	$1.72 \mu_B$
Ni	$0.60 \mu_B$
Gd	$7.12 \mu_B$

$$\chi_m \uparrow\uparrow (\text{to } 10^6) \rightarrow M \gg H \rightarrow B \cong \mu_0 M$$

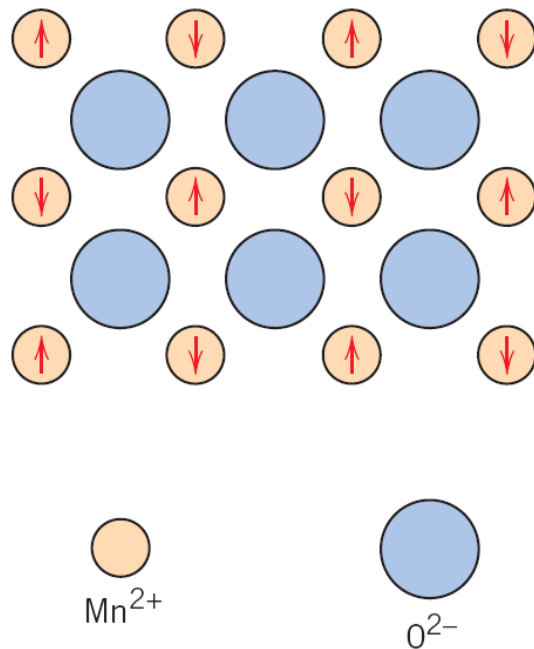
The permanent magnetic moment of the ferromagnetic materials is due fundamentally to the spin magnetic moments, although there is a small contribution from the orbital magnetic moments

Iron, Nickel, Cobalt

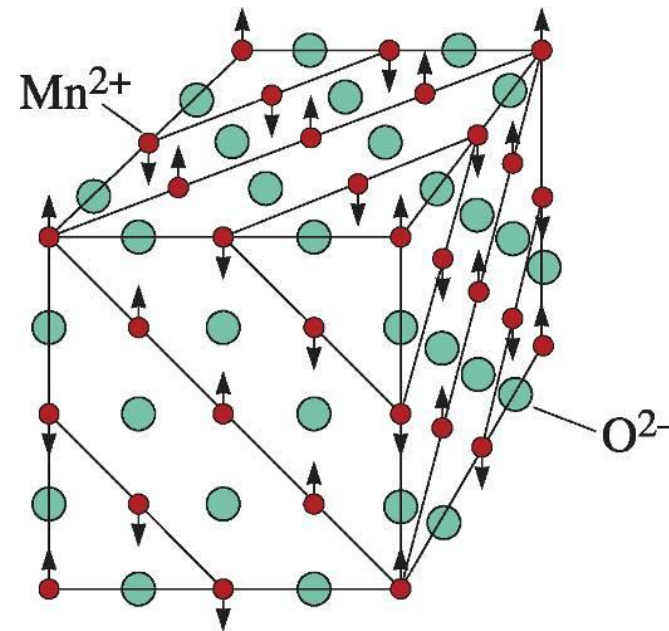
ANTI-FERROMAGNETISM

It is present in materials with permanent magnetic dipoles oriented in anti-parallel way. The solid as a whole has no net magnetic moment

Antiparallel alignment of spin magnetic moments for antiferromagnetic manganese oxide.

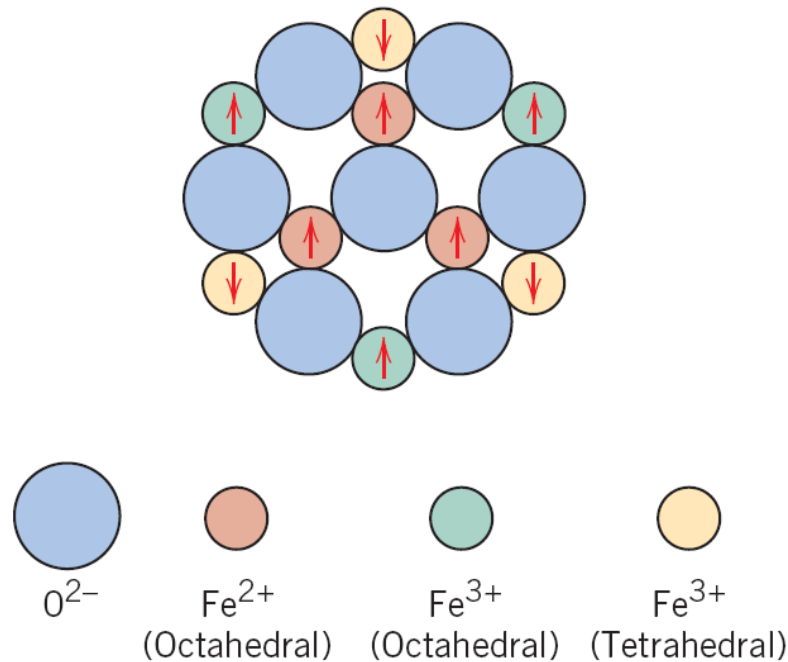


Examples: Cr, Mn, MnO



FERRIMAGNETISM

It is present in ceramic materials with permanent magnetic dipoles of different intensity, aligned in anti-parallel way.



<i>Cation</i>	<i>Net Spin Magnetic Moment (Bohr magnetons)</i>
Fe ³⁺	5
Fe ²⁺	4
Mn ²⁺	5
Co ²⁺	3
Ni ²⁺	2
Cu ²⁺	1

Ferrimagnetic materials



Ceramics: Ferrites (chemical compounds with the formula AB₂O₄), Garnets (M₃Fe₅O₁₂), AB₁₂O₁₉

FERRIMAGNETISM

SOME COMMERCIAL FERRITES COMPOSITIONS

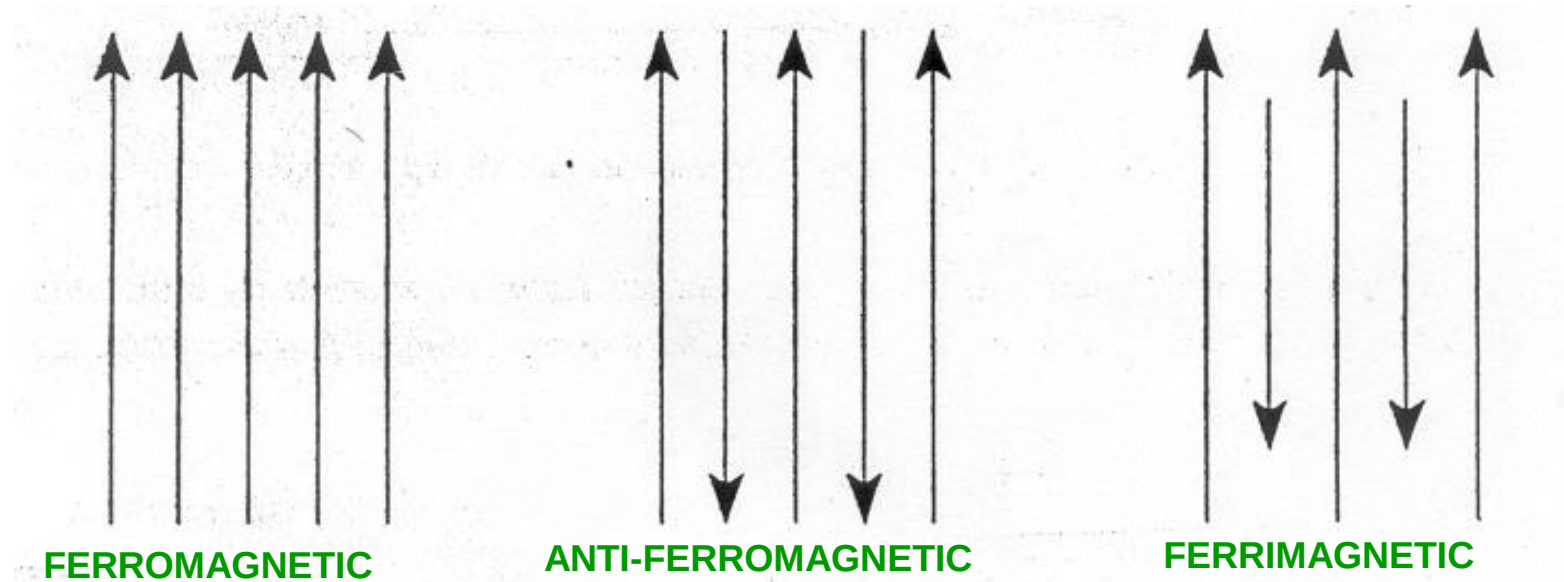
Name	Composition	Comments
Magnesium ferrite	MgFe_2O_4	
Magnesium–zinc ferrite	$\text{Mg}_x\text{Zn}_{1-x}\text{Fe}_2\text{O}_4$	$0 < x < 1.$
Manganese ferrite	MnFe_2O_4	
Manganese–iron ferrite	$\text{Mn}_x\text{Fe}_{3-x}\text{O}_4$	$0 < x < 3.$
Manganese–zinc ferrite	$\text{Mn}_x\text{Zn}_{1-x}\text{Fe}_2\text{O}_4$	$0 < x < 1.$
Nickel ferrite	NiFe_2O_4	
Lithium ferrite	$\text{Li}_{0.5}\text{Fe}_{2.5}\text{O}_4$	Li^+ occurs in combination with Fe^{3+} , producing a $(\text{Li}_{0.5}\text{Fe}_{0.5})^{2+}\text{Fe}_2^{3+}\text{O}_4$ configuration.

SOME COMMERCIAL GARNET COMPOSITIONS

Name	Composition	Comments
Yttrium iron garnet (YIG)	$\text{Y}_3\text{Fe}_5\text{O}_{12}$	
Aluminum substituted YIG	$\text{Y}_3\text{Al}_x\text{Fe}_{5-x}\text{O}_{12}$	Al^{3+} prefers tetrahedral sites.
Chromium substituted YIG	$\text{Y}_3\text{Cr}_x\text{Fe}_{5-x}\text{O}_{12}$	Cr^{3+} prefers octahedral sites.
Lanthanum iron garnet (LaIG)	$\text{La}_3\text{Fe}_5\text{O}_{12}$	
Praseodymium iron garnet (PrIG)	$\text{Pr}_3\text{Fe}_5\text{O}_{12}$	Many of the pure garnets are not commercially prepared; all form at least limited solid solutions with each other (e.g., $\text{Pr}_x\text{Y}_{3-x}\text{Fe}_5\text{O}_{12}$ with $x_{\max} = 1.5$).

SUMMARY OF TYPES OF MAGNETISM

ORIENTED DIPOLES IN THE ABSENCE OF MAGNETIC FIELD

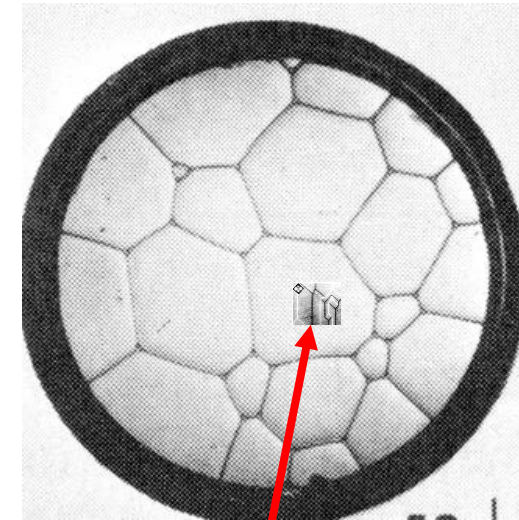
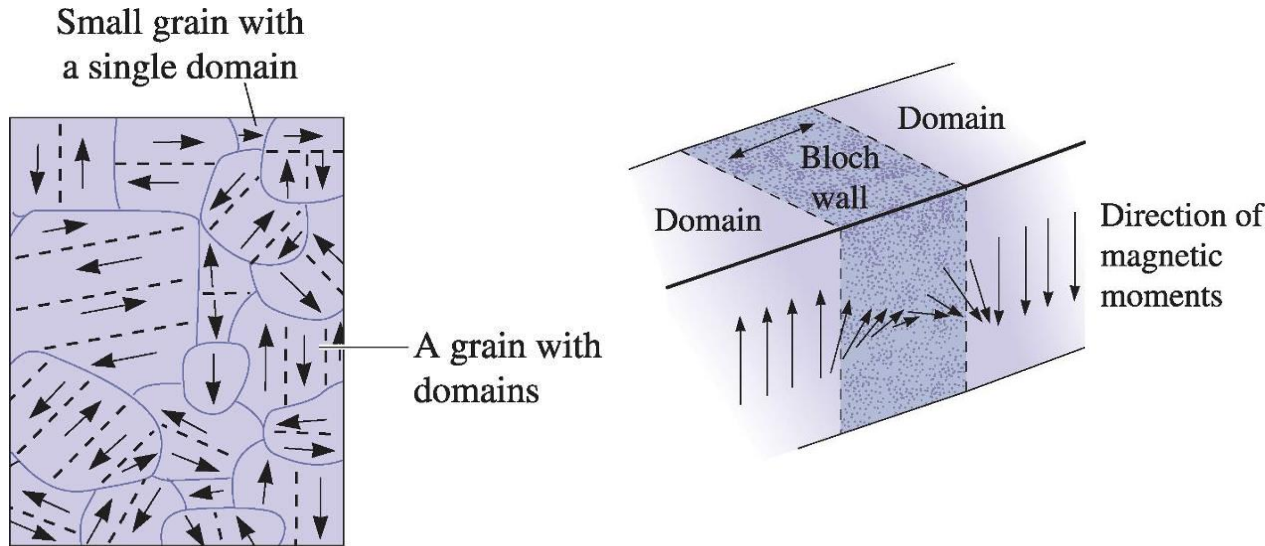


DIPOLES ORIENTED ONLY IN THE PRESENCE OF A MAGNETIC FIELD PARAMAGNETIC

WITHOUT PERMANENT DIPOLES, ONLY INDUCED DIPOLES DIAMAGNETIC

MAGNETIC DOMAINS AND BLOCH WALL

A **magnetic domain** is a tridimensional region within a magnetic (ferromagnetic or ferrimagnetic) material with uniform magnetization



Domains in iron

(001)

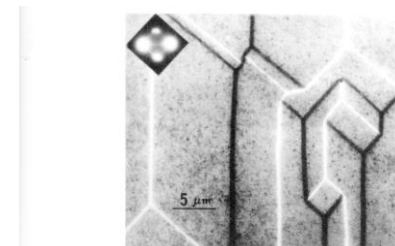


Fig. 15.16. Domain pattern observed in a (001) iron single crystal film of thickness 2500 Å, and its diffraction pattern (inset).¹⁹

Domain are usually smaller than grains

A **Bloch wall** is a narrow transition region at the **boundary between magnetic domains**, over which the **magnetization progressively changes** from its value in one domain to that in the next

Dimension of the domains $\approx 10 - 100 \mu\text{m}$

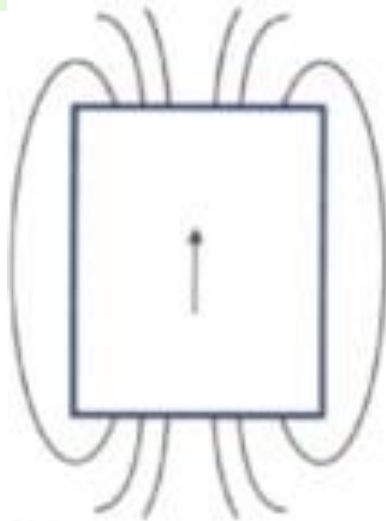
Dimension of the walls $\approx 100 \text{ nm}$ (300 atoms approximately)

The existence of magnetic domains makes **ferromagnetic and ferrimagnetic** materials which have *never* been exposed to a magnetic field have a null net magnetization

ORIGIN OF MAGNETIC DOMAINS

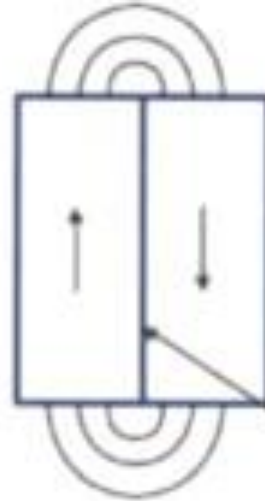
The magnetic domain structure is generated spontaneously to decrease the magnetostatic energy of the material

High magnetostatic energy



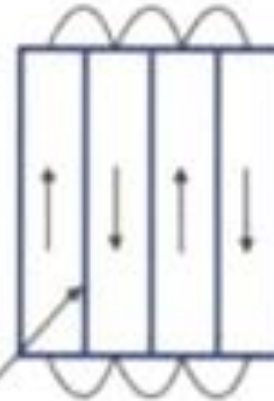
A

Lower magnetostatic energy than (a)



B

Lower magnetostatic energy than (b)



C

zero magnetostatic energy



D

Magnetic domain walls

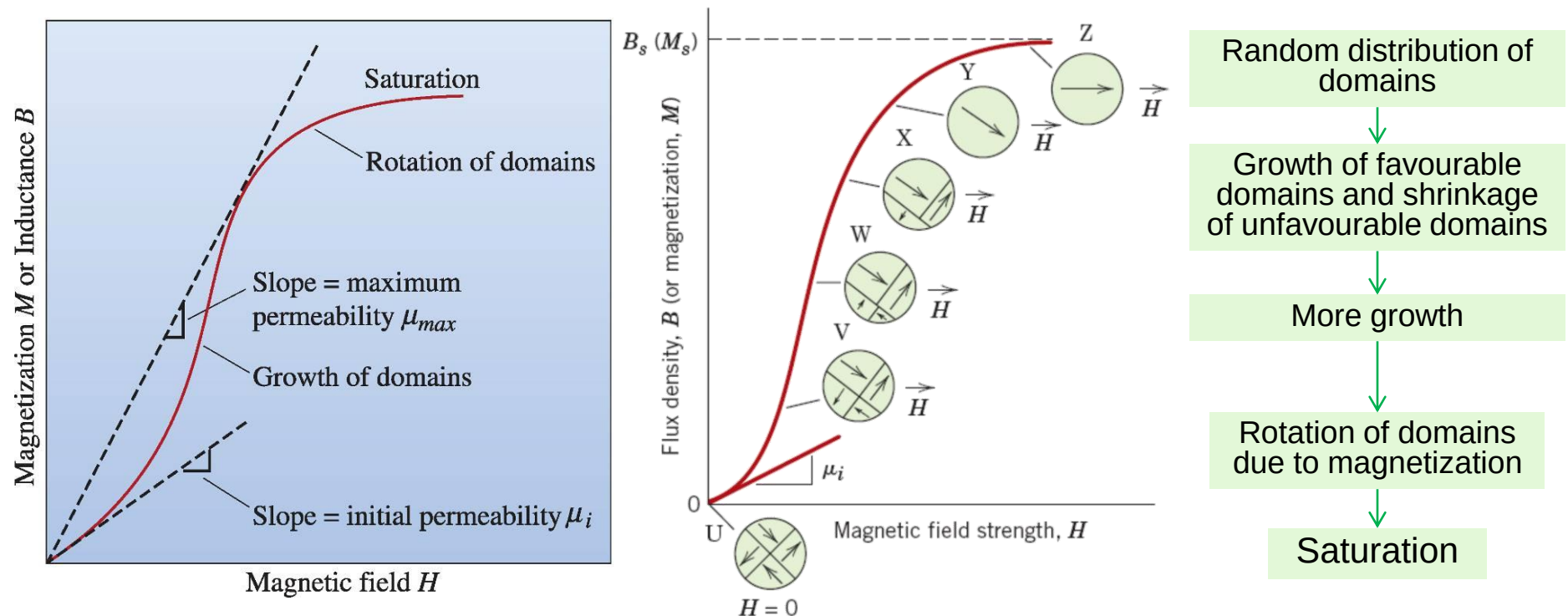
One domain (a)

Two domains (b)

Four domains (c)

EVOLUTION OF MAGNETIZATION DUE TO H

When a magnetic field is applied to a magnetic material: magnetization initially increases slowly, then more rapidly as the domain begin to grow. Later the magnetization slows as domain must eventually rotate to reach saturation.



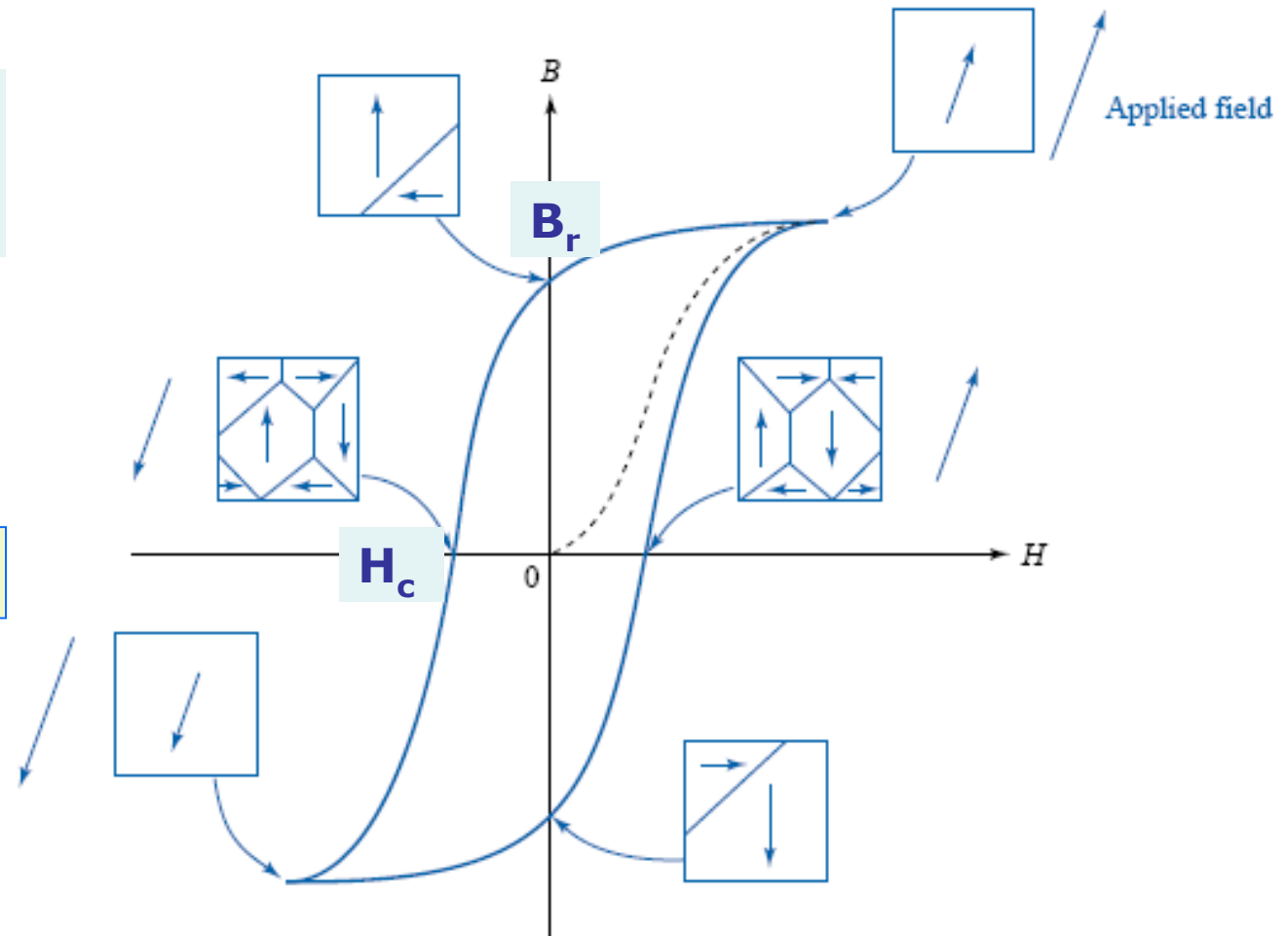
M_s = **saturation magnetization**: magnetization that results when all the magnetic dipoles in a solid are aligned with the external field (saturation flux density B_s)

μ_i = initial permeability
 μ_{max} = maximum permeability
 (μ varies with respect to B)

HYSTERESIS CYCLES

 B_r = remanence H_c = coercivity

1/(how easy is to change in
orientation of the dipoles)



TYPES OF MATERIALS ACCORDING TO THEIR HYSTERESIS CYCLES

- Soft magnetic materials

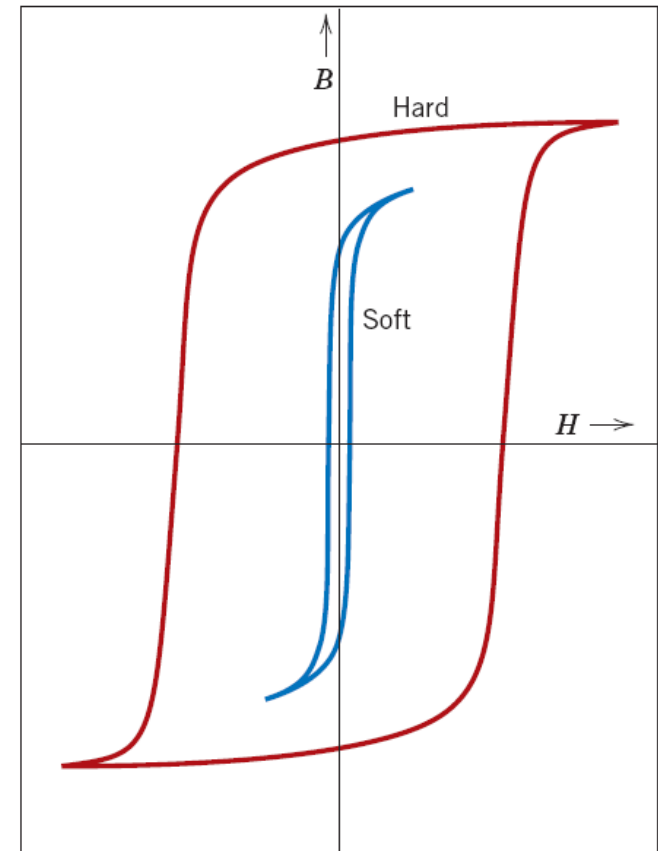
(Bloch walls are easy to move)
High initial permeability and low H_c

Materials under alternating electrical fields (in order to increase B and produce fast flux changes in coils and circuits with minimum energy losses)

-Hard magnetic materials

(Bloch walls are difficult to move)
High remanence, B_s and H_c ; Low μ_i

Permanent magnets
(because of their high resistance to demagnetization)



Límit: **$H_c = 10.000 \text{ A/m}$**

APPLICATIONS OF MAGNETIC MATERIALS ACCORDING TO THEIR HYSTERESIS CYCLES

Permanent magnets

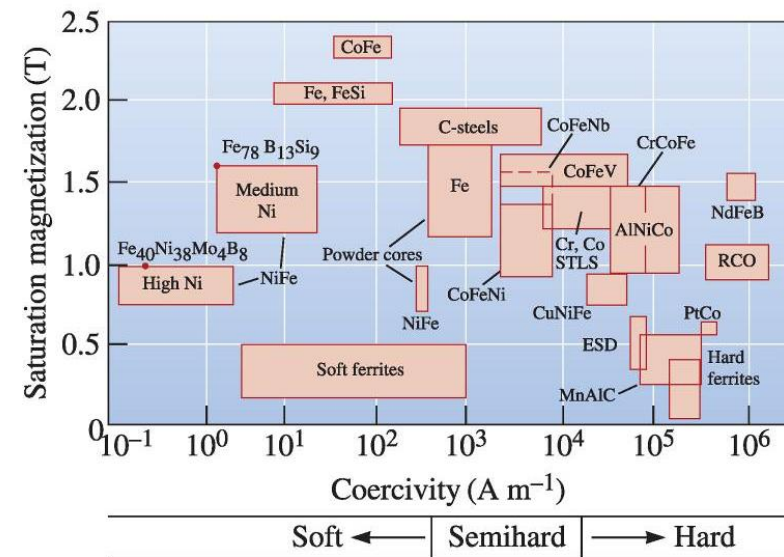
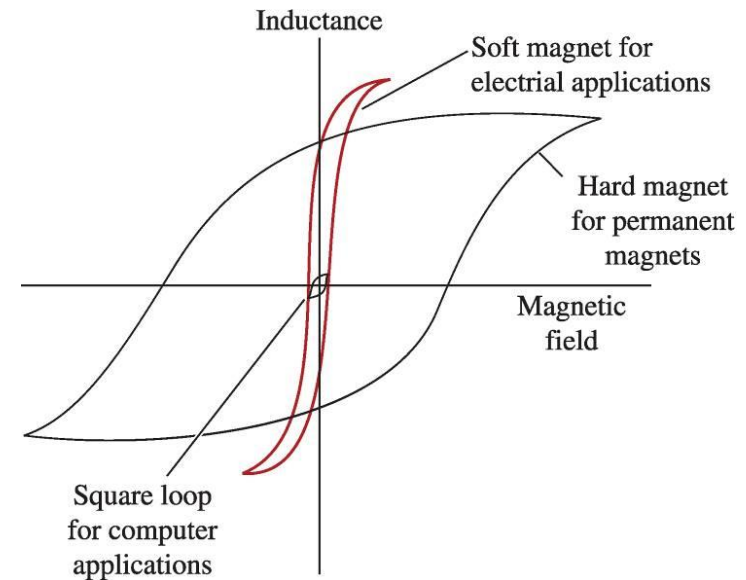
Hard magnetic materials, with high resistance to demagnetization: high remanance, high coercive field

Materials under alternating electrical fields (in order to increase magnetic flux in coils, circuits, cores for elctromagnets, transformers, generators, electric motors)

Soft magnetic materials, to minimize energy losses: high saturation magnetization, small coercive field, high permeability

Magnetic memories in computers, magnetic amplifiers, pulse transformers (data storage)

Rectangular hysteresis cycle materials: low remanance, low saturation magnetization, low coercive filed (soft magnetic materials modified by thermal treatment in the presence of a magnetic field)



(b)

FACTORS AFFECTING THE HYSTERESIS CYCLE

- The number and intensity of its dipoles
 - The type of magnetism of the material
 - Its microstructure: (grain size, pores, impurities, etc)
 - Presence of strains due to mechanical deformations
 - Working temperature
 - Radiation
 - Electrical currents
- } **Material's composition**
 } **Processing**
 } **Working conditions**

➤ **Coercivity**: strongly microstructure-dependant property (crystallographic direction, grain boundaries, pores, shape of particles)

➤ **Saturation magnetization** : constant for a material of a given compositions

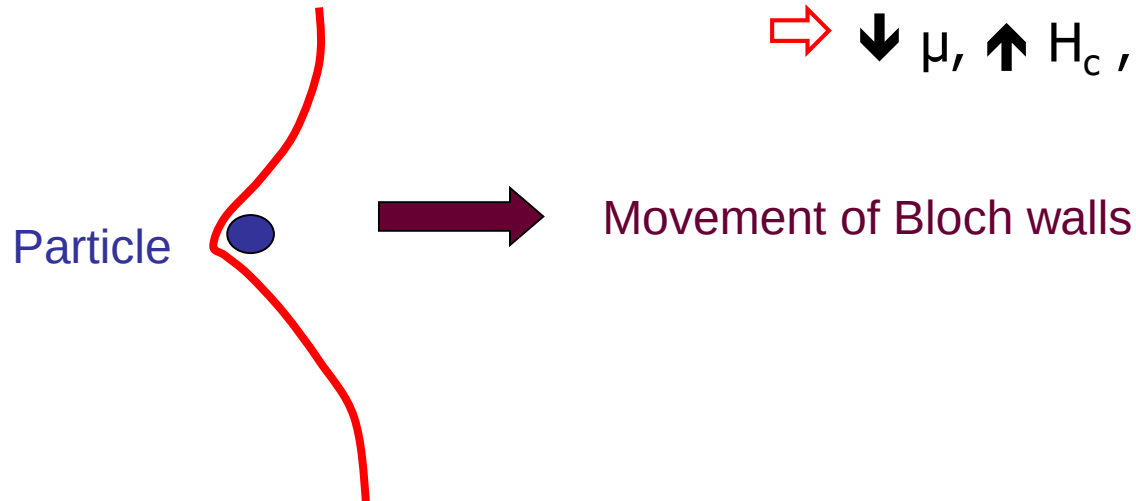
EFFECT OF THE MICROSTRUCTURE IN THE HYSTERESIS CYCLE

↑ Size of grain $\Rightarrow$ ↑ Mobility of Bloch walls

$\Rightarrow$ ↑ μ , ↓ H_c , ↓ B_r

↑ Inclusions, dislocations, pores, etc. $\Rightarrow$ ↓ Mobility of Bloch walls

$\Rightarrow$ ↓ μ , ↑ H_c , ↑ B_r



EFFECT OF TEMPERATURE IN THE HYSTERESIS CYCLE

When temperature $\uparrow \Rightarrow \uparrow$ mobility of dipoles

The maximum dipole alignment achieved when the field is applied is **less** than before.

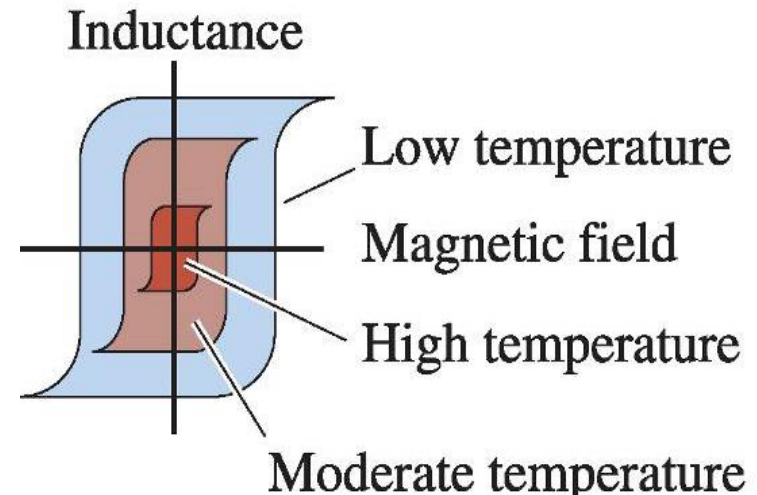
M_s is reduced

Makes difficult to keep the dipoles aligned when the field disappears

B_r and H_c are reduced

When $T > T_c$ dipoles can still align with H , but when H disappears, their alignments become random

The material becomes paramagnetic



TEMPERATURES AND MAGNETIC BEHAVIOUR

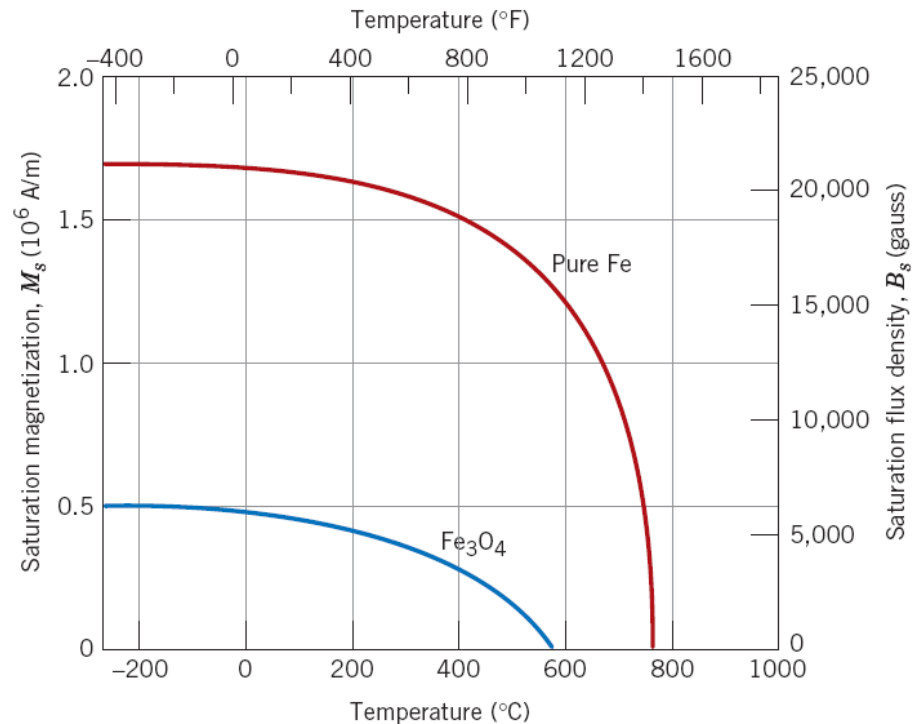
Thermal agitation tends to misalign dipoles

Curie Temperature

T_c = Temperature at which the saturation magnetization abruptly descends to 0

For $T > T_c$, ferromagnetic and ferrimagnetic materials become paramagnetic

Ferromagnetic Material	T_c (°K)
Fe	1043
Co	1404
Ni	631
Gd	289



Néel temperature

Temperature at which an antiferromagnetic material becomes paramagnetic – that is, the thermal energy becomes large enough to destroy the macroscopic magnetic ordering within the material.

Material antiferromagnetic	Néel temperature (°K)
α -Mn	100
MnO	116
Cr	310
NiO	523

TYPICAL PROPERTIES OF MAGNETIC MATERIALS

TABLE 20-4 ■ Properties of selected soft magnetic materials

Name	Composition	Permeability μ_r		Coercivity H_c (A · m ⁻¹)	Retentivity B_r (T)	$B_{\max}$ (T)	Resistivity ($\mu\Omega \cdot \text{m}$)
		Initial	Maximum				
Ingot Iron	99.8% Fe	150	5000	80	0.77	2.14	0.10
Low-carbon steel	99.5% Fe	200	4000	100		2.14	1.12
Silicon iron, unoriented	Fe-3% Si	270	8000	60		2.01	0.47
Silicon iron, grain-oriented	Fe-3% Si	1400	50,000	7	1.20	2.01	0.50
4750 alloy	Fe-48% Ni	11,000	80,000	2		1.55	0.48
4-79 permalloy	Fe-4% Mo-79% Ni	40,000	200,000	1		0.80	0.58
Superalloy	Fe-5% Mo-80% Ni	80,000	450,000	0.4		0.78	0.65
2V-Permendur	Fe-2% V-49% Co	800	450,000	0.4		0.78	0.65
Supremendur	Fe-2% V-49% Co		100,000	16	2.00	2.30	0.40
Metglas ^a 2650SC	Fe ₈₁ B _{13.5} Si _{3.5} C ₂		300,000	3	1.46	1.61	1.35
Metglas ^a 2650S-2	Be ₇₈ B ₁₃ S ₉		600,000	2	1.35	1.56	1.37
MnZn Ferrite	H5C2 ^b	10,000		7	0.09	0.40	1.5×10^5
MnZn Ferrite	H5E ^b	18,000		3	0.12	0.44	5×10^4
NiZn Ferrite	K5 ^b	290		80	0.25	0.33	2×10^{12}

TABLE 20-6 ■ Properties of selected hard, or permanent, or magnetic materials

Material	Common Name	$\mu_0 M_r$ (T)	$\mu_0 H_c$ (T)	$(BH)_{\max}$ (kJ · m ⁻³)
Fe-Co	Co-steel	1.07	0.02	6
Fe-Co-Al-Ni	Alnico-5	1.05	0.06	44
BaFe ₁₂ O ₁₉	Ferrite	0.42	0.31	34
SmCo ₅	Sm-Co	0.87	0.80	144
Nd ₂ Fe ₁₄ B	Nd-Fe-B	1.23	1.21	290–445

TABLE 20-5 ■ Properties of typical magnetic recording materials in a powder form

	Particle Length (μm)	Aspect Ratio	Magnetization B_r		Coercivity H_c		Surface Area (m ² /g)	Curie Temp. T_c (°C)
			(Wb/m ²)	(emu/cm ³)	(kA/m)	(Oe)		
γ -Fe ₂ O ₃	0.20	5:1	0.44	350	22–34	420	15–30	600
Co- γ -Fe ₂ O ₃	0.20	6:1	0.48	380	30–75	940	20–35	700
CrO ₂	0.20	10:1	0.50	400	30–75	950	18–55	125
Fe	0.15	10:1	1.40 ^a	1100 ^a	56–176	2200	20–60	770
Barium Ferrite	0.05	0.02 μm	0.40	320	56–240	3000	20–25	350

SUPERCONDUCTIVITY AND MAGNETISM

Superconductivity

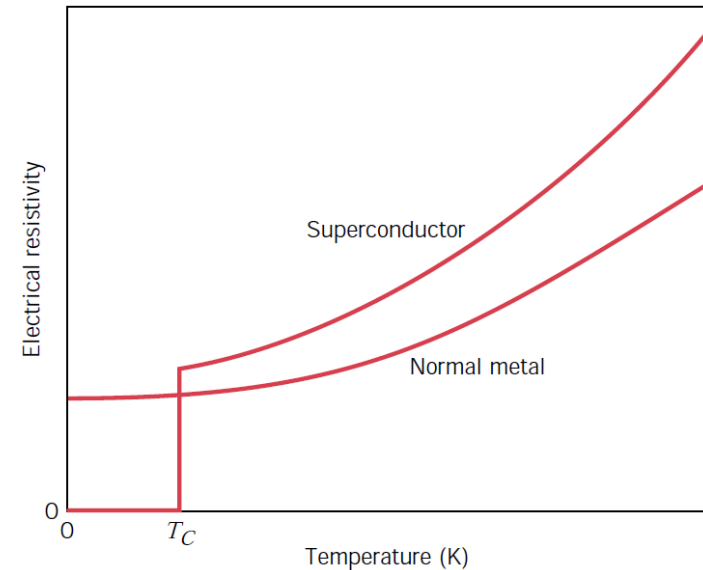
The **electrical resistivity** of a superconductor becomes **zero** below some **critical temperature T_c**

Superconductors can be:

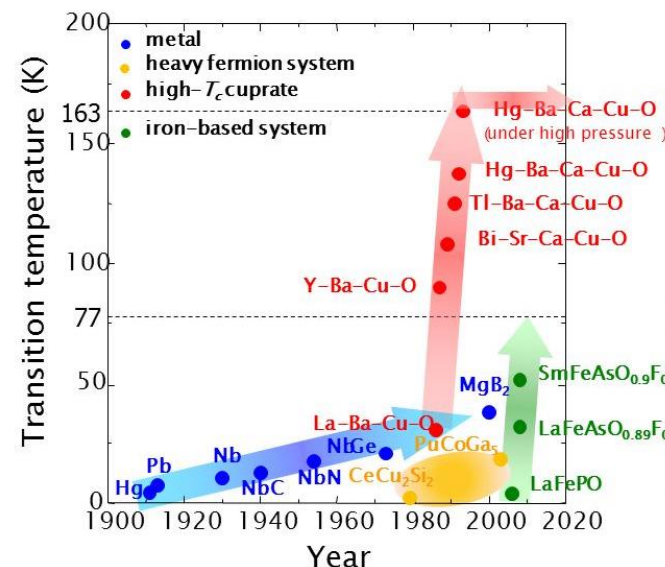
- Pure metal and alloys
- Ceramics (typically with copper oxide)
- Organic compounds (based, for example, in Se or S)

T_c ranges between 0.01 and 165 K (*limitation*)

Usually, T_c are lower for metals than for ceramics



Material	T_c (K)	Material	T_c (K)
Al	1.20	Nb ₃ Sn	18.05
Hg	4.15	Nb ₃ Al	17.5
Mo	0.92	V ₃ Ga	16.5
Nb	9.26	La ₃ In	10.4
Pb	7.19	LaBaCuO	35
Ta	4.48	YBCO	90
Ti	0.39	Tl ₂ Ba ₂ Ca ₂ Cu ₃ O ₁₀	125
Zn	0.88	Hg ₁₂ Tl ₃ Ba ₃₀ Ca ₃₀ Cu ₄₅ O ₁₂₅	138



1911

Discovery of superconducting phenomenon

1979

Heavy fermion superconductor

1986

High- T_c cuprate superconductor

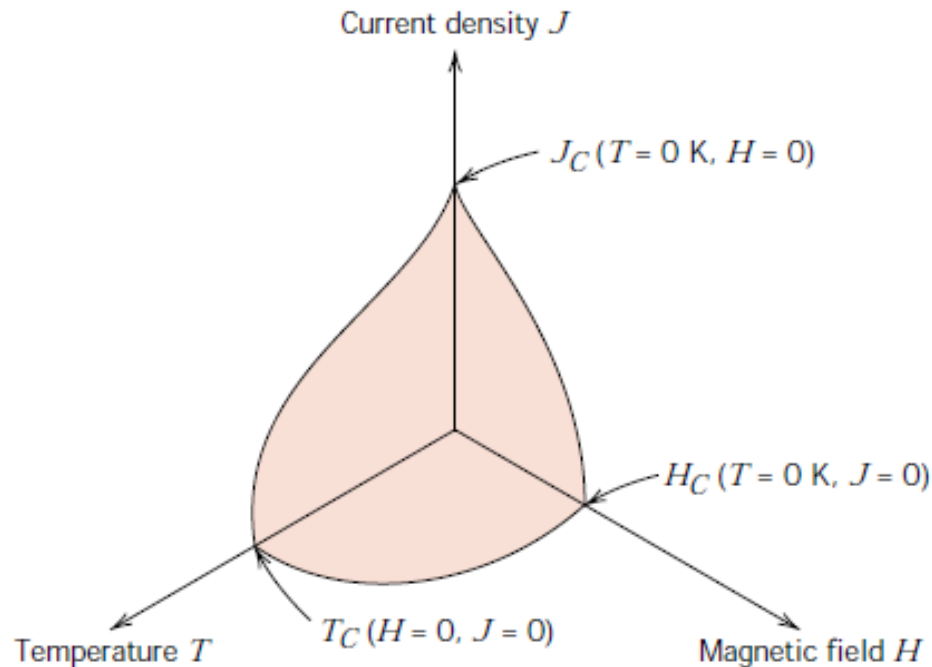
2006

Iron-based high- T_c superconductor

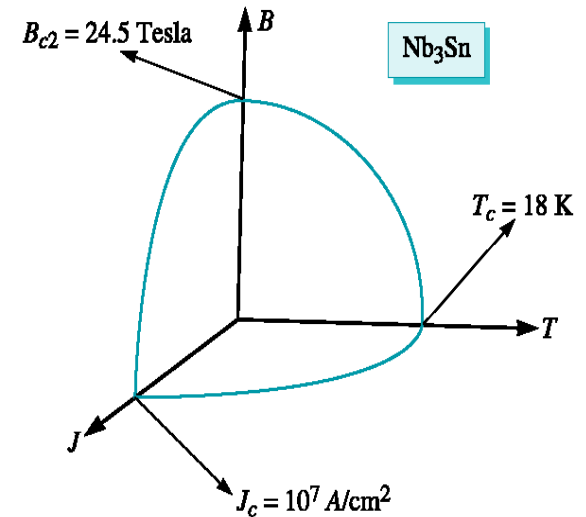
SUPERCONDUCTIVITY AND MAGNETISM

Critical Magnetic Field (H_c) and Critical Current Density (J_c)

At temperatures below T_c , the superconducting state will stop upon application of a sufficiently large magnetic field (H_c) or current density (J_c)



H_c and J_c decreases with increasing temperature

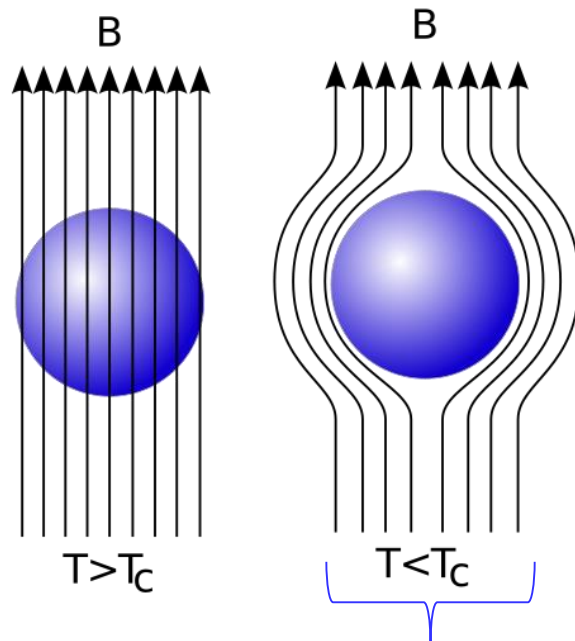


The superconducting envelope or surface showing the combined effects of temperature, magnetic field, and current density on a Nb₃Sn superconductor.

SUPERCONDUCTIVITY AND MAGNETISM

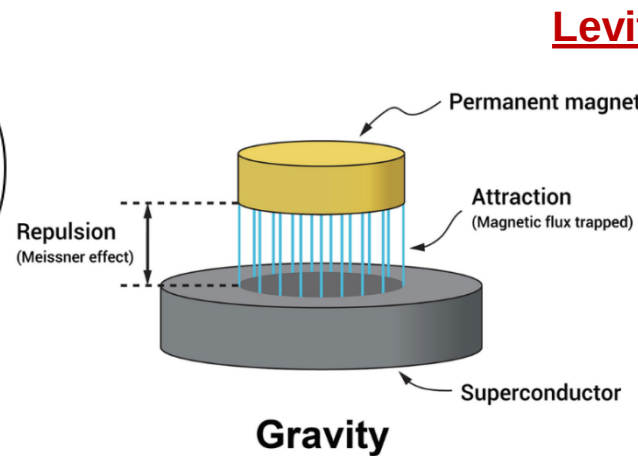
Meissner effect

The Meissner-Oschensfeld effect consists in the expulsion of the magnetic field from a material when it is in semiconductor state

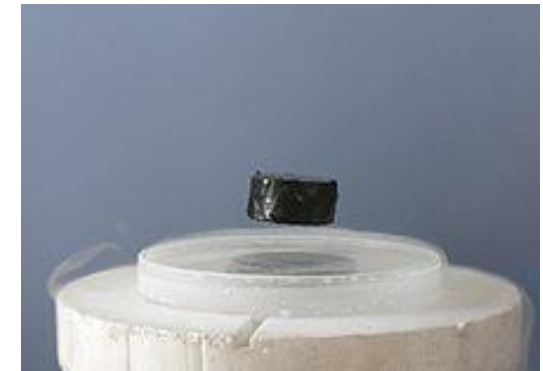


The material becomes
diamagnetic

Superconductors in the Meissner state exhibit **perfect diamagnetism**, or superdiamagnetism, meaning that the total magnetic field is very close to zero deep inside



Levitation

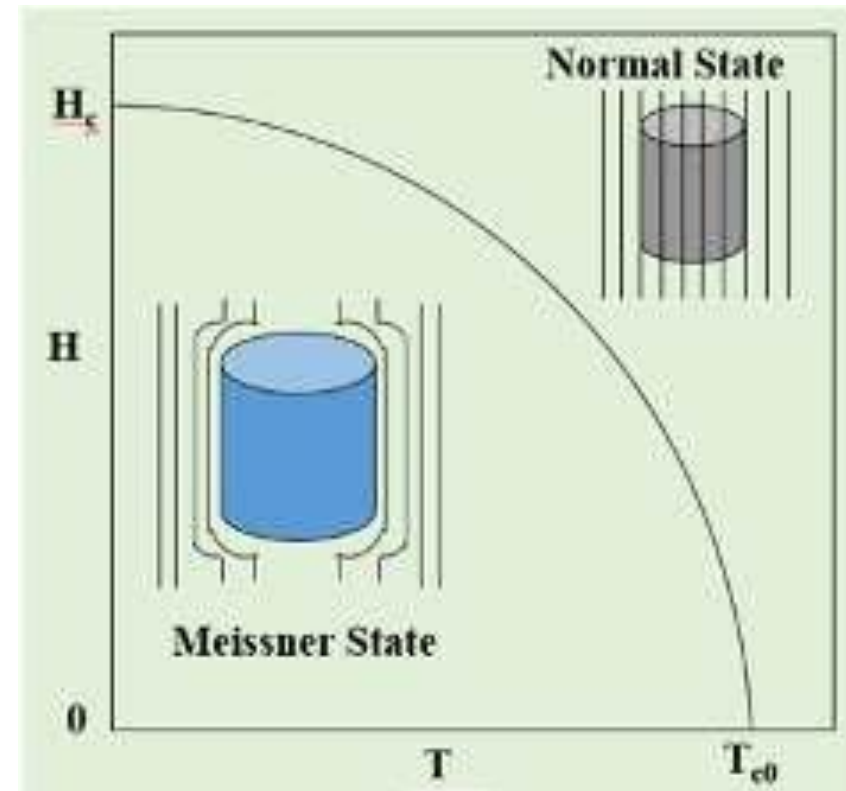
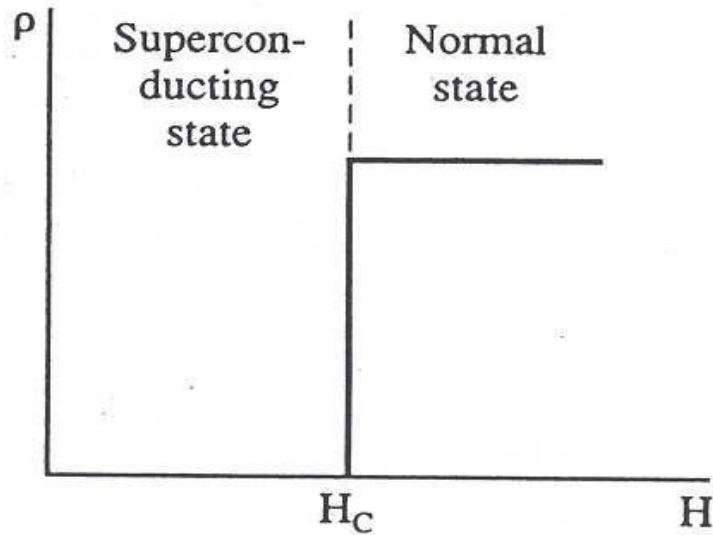


A magnet levitating above
a superconductor cooled
by liquid nitrogen

TYPES OF SUPERCONDUCTORS

Type I (soft superconductors):

Sharp transition to normal state. It takes place at very low $H_c \Rightarrow$ Low practical interest

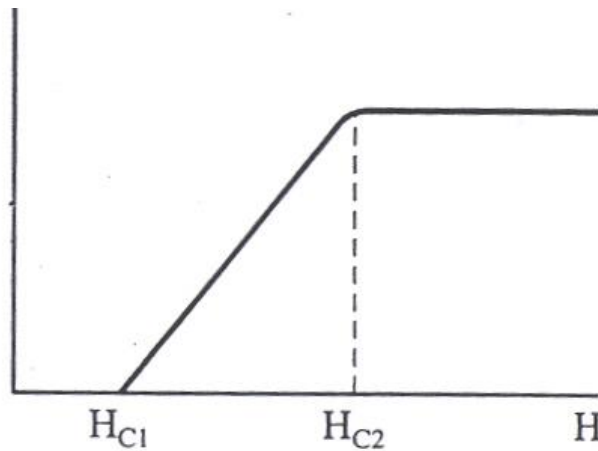


Type I semiconductors: Pure metals (Sn, Al, Pb and Hg)

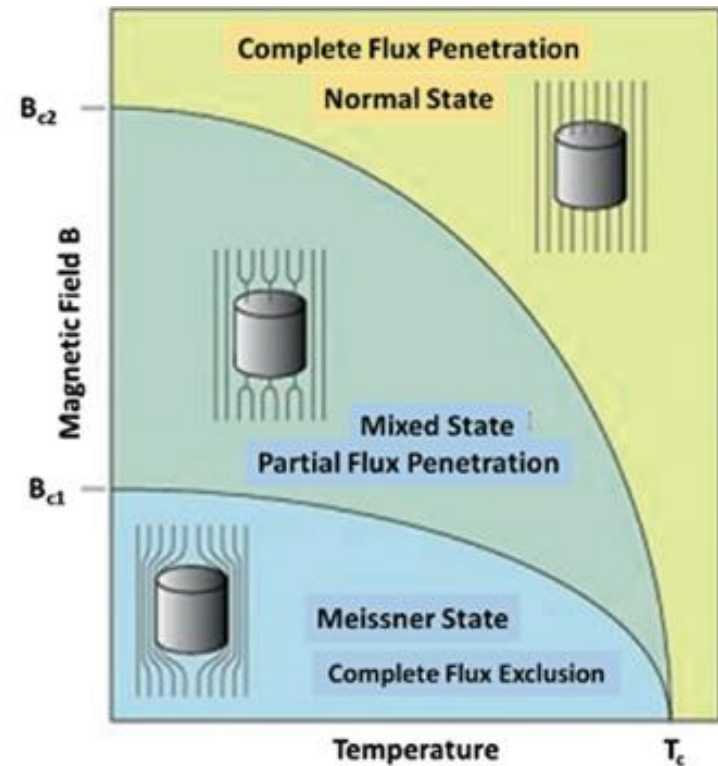
TYPES OF SUPERCONDUCTORS

Type II (hard superconductors):

Gradual transition to normal state with H (the lowest critical field, H_{C1} , can be 100 times lower than the highest critical field, H_{C2}) $\Rightarrow$ **High resistance against the destruction of superconductivity by magnetic fields**



The magnetic flux lines begin to penetrate the material at H_{C1} , and at H_{C2} , field penetration is complete. For fields between the material exists in a mixed state both normal and superconducting regions are present.



Type II semiconductors: - alloys (i.e. Nb-Zr, Nb-Ti) and intermetallics (i.e. Nb₃Sn)
- ceramics (YBaCuO)

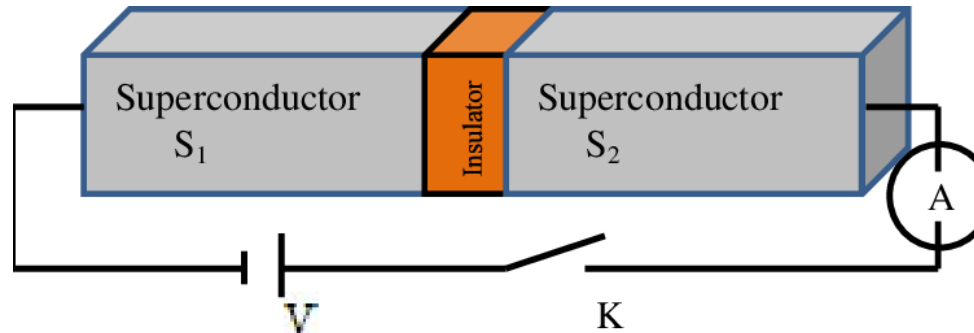
SUPERCONDUCTIVITY AND MAGNETISM

<https://www.youtube.com/shorts/Bo8WV032dyk>



SUPERCONDUCTIVITY

Josephson Effect



The Josephson or Tunnel effect occurs when a current flows through two superconductors separated by a very thin isolating layer.

Tunnel Junction: Very thin oxide layer (≈ 3 nm thickness)

Josephson current: - It goes through the tunnel junctions though there is no potential difference

- A D.C. voltage applied through the junction causes a RF current oscillation (and a RF voltage through the junction causes a DC current)

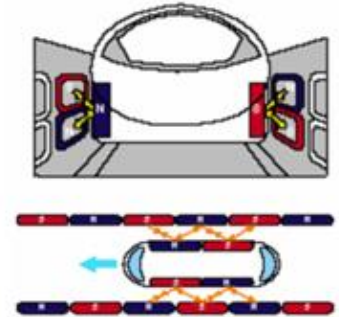
- Its value is very sensitive to the magnetic field



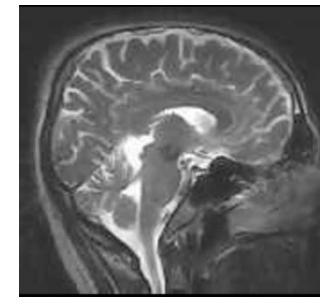
Many applications

APPLICATIONS OF SUPERCONDUCTORS

- Levitated transport



- Powerful superconducting electromagnets are used in magnetic resonance imaging (MRI).
- RF and microwave filters (e.g., for mobile phone base stations, as well as military ultra-sensitive/selective receivers)
- Large hadron collider or particle accelerator.
- Very low electrical-loss components, known as filters, based on ceramic superconductors, have been developed for wireless communications.

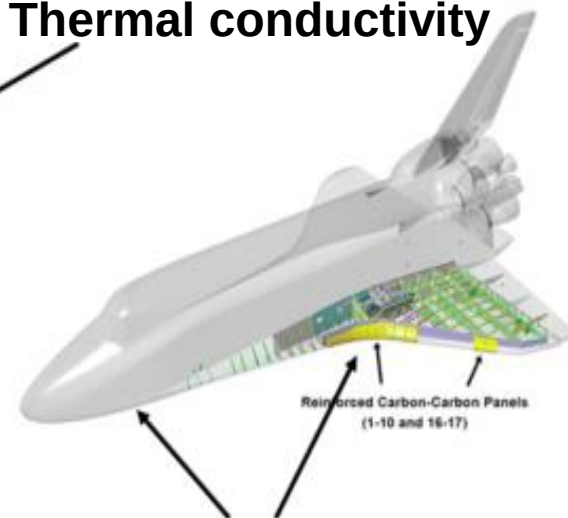


THERMAL PROPERTIES

- Introduction
- Heat Capacity and Specific Heat
- Thermal Expansion
- Thermal Conductivity
- Thermal Shock



Thermal conductivity



HEAT CAPACITY AND SPECIFIC HEAT

Thermal energy in solids: **increase in vibrational energy** of atoms

⇒ vibrations produce traveling lattice waves

Phonon - A packet of elastic waves (thermal vibrational energy).

Energy of a phonon: $E = \frac{h\nu}{\lambda} = h\omega$

h = Planck constant
ν = speed of sound
λ = wavelength
ω = frequency

Heat Capacity: Energy required to raise the temperature of **one mole of a material one degree** (material's ability to absorb heat)

$$C = \frac{dQ}{dT}$$

C = heat capacity (J/mol-K, or cal/mol-K)
dQ = energy required to produce a dT temperature change

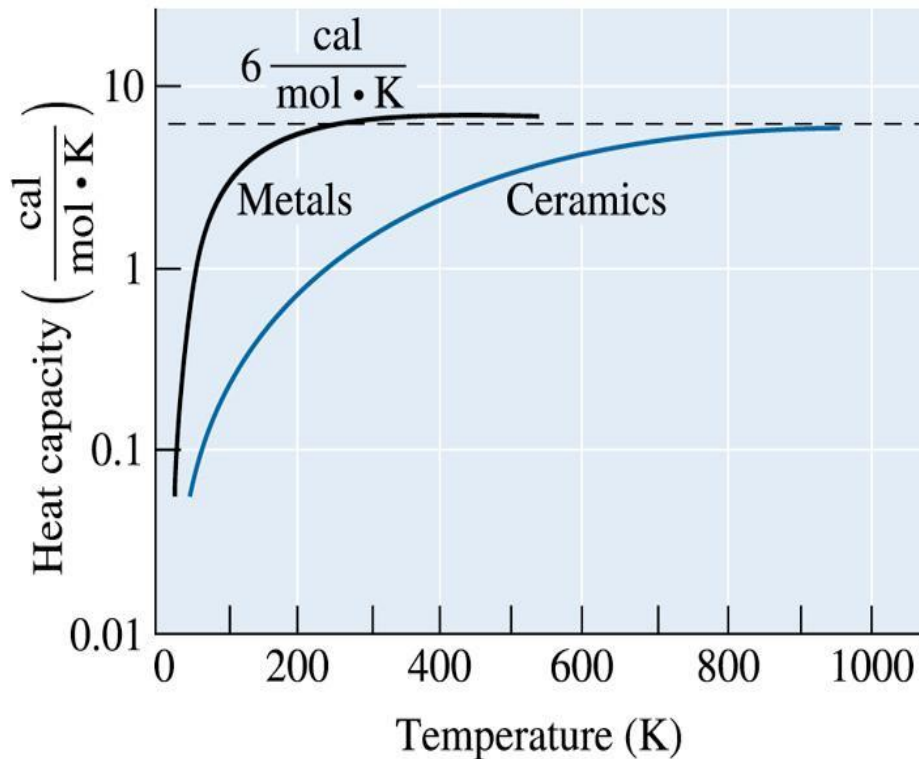
Specific heat - The energy required to **raise the temperature of one gram of a material by one degree**.

$$\text{Specific Heat} = C_p = \frac{\text{heat capacity}}{\text{atomic weight}}$$

HEAT CAPACITY AND SPECIFIC HEAT

Heat capacity **increases with temperature** $\Rightarrow$ increased ability of the lattice waves to enhance their average energy.

Above the **Debye temperature** θ_D , the heat capacity becomes *independent of temperature* (approximately $3R \approx 6 \text{ Cal/mol.K}$, R = gas constant)



Heat capacity and specific heat for most materials:

❑ Does NOT depend significantly on the structure of the material (dislocations, vacancies, grain size)

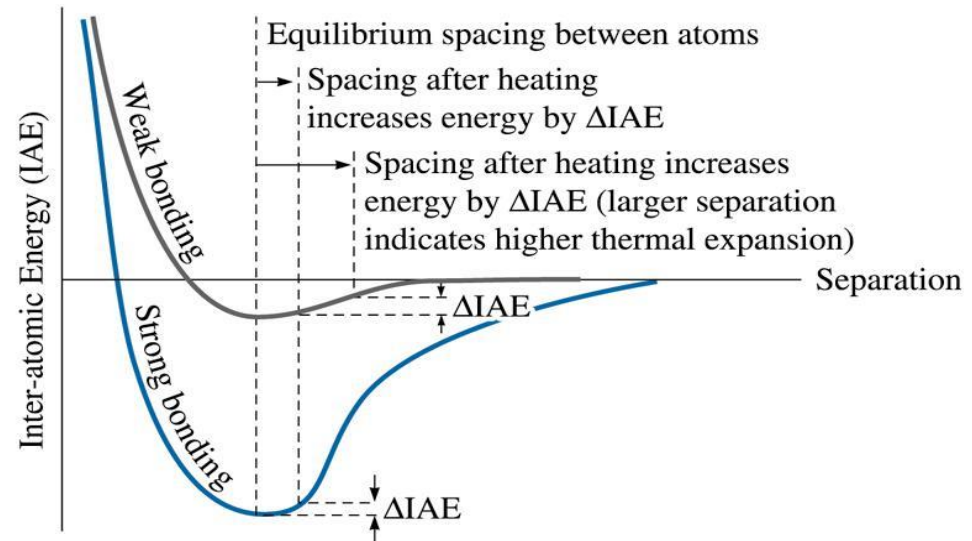
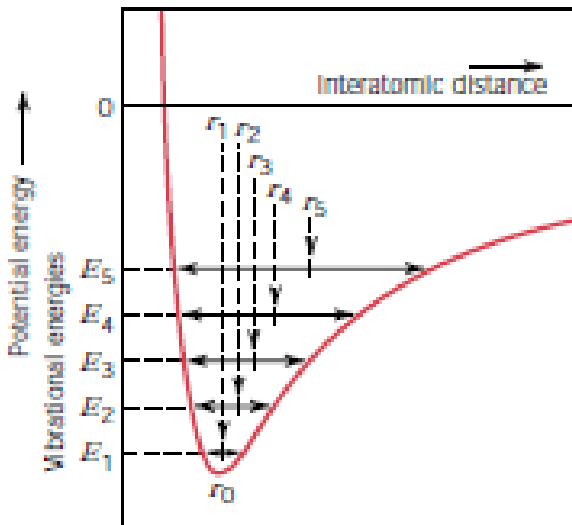
➤ *Except: **ferromagnetic materials** (Iron): high specific heat at the curie temperature (change from ferromagnetic to paramagnetic)*

Heat capacity as a function of temperature for metals and ceramics

THERMAL EXPANSION

Linear coefficient of thermal expansion α - amount by the unit length of a material changes when the temperature of the material changes

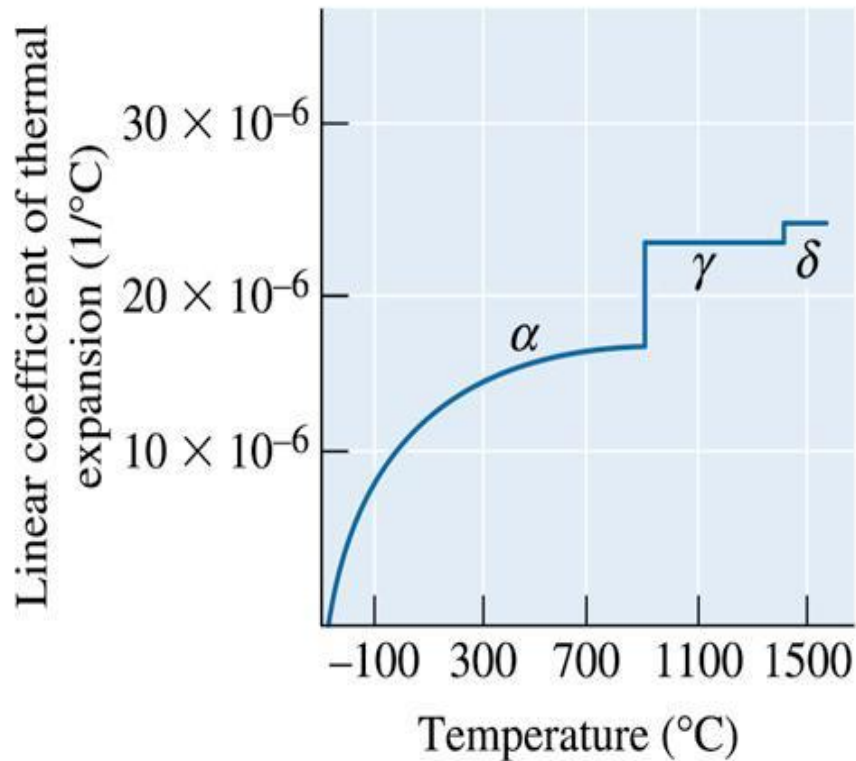
Thermal expansion
$$\frac{l_f - l_0}{l_0} = \alpha(T_f - T_0) \quad \frac{\Delta l}{l_0} = \alpha \Delta T$$



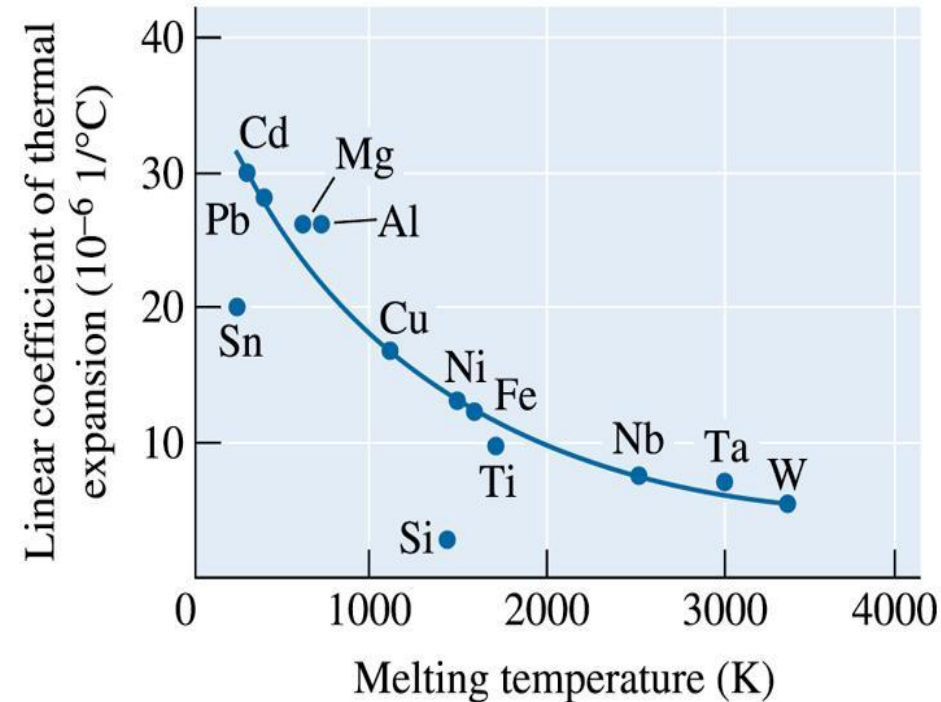
- ❑ Materials that display a **high bonding energy** (strong atomic bonding) have **low coefficients of thermal expansion**.

THERMAL EXPANSION

❑ **Allotropic materials** have abrupt changes in their dimensions when phase transformations occurs (cracking in refractories)



The linear coefficient of thermal expansion of iron changes abruptly at temperatures where an allotropic transformation occurs



The relationship between the linear coefficient of thermal expansion and the melting temperature in metals at 25°C

THERMAL EXPANSION

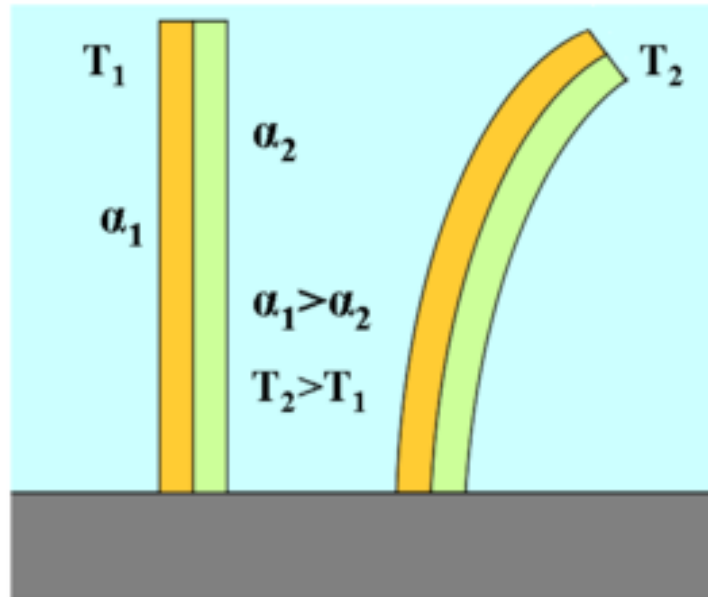
<i>Material</i>	c_p (J/kg·K) ^a	α_l [(°C) ⁻¹ × 10 ⁻⁶] ^b	k (W/m·K) ^c
<i>Metals</i>			
Aluminum	900	23.6	247
Copper	386	17.0	398
Gold	128	14.2	315
Iron	448	11.8	80
Nickel	443	13.3	90
Silver	235	19.7	428
Tungsten	138	4.5	178
1025 Steel	486	12.0	51.9
316 Stainless steel	502	16.0	15.9
Brass (70Cu-30Zn)	375	20.0	120
Kovar (54Fe-29Ni-17Co)	460	5.1	17
Invar (64Fe-36Ni)	500	1.6	10
Super Invar (63Fe-32Ni-5Co)	500	0.72	10
<i>Ceramics</i>			
Alumina (Al ₂ O ₃)	775	7.6	39
Magnesia (MgO)	940	13.5 ^d	37.7
Spinel (MgAl ₂ O ₄)	790	7.6 ^d	15.0 ^e
Fused silica (SiO ₂)	740	0.4	1.4
Soda-lime glass	840	9.0	1.7
Borosilicate (Pyrex) glass	850	3.3	1.4
<i>Polymers</i>			
Polyethylene (high density)	1850	106–198	0.46–0.50
Polypropylene	1925	145–180	0.12
Polystyrene	1170	90–150	0.13
Polytetrafluoroethylene (Teflon)	1050	126–216	0.25
Phenol-formaldehyde, phenolic (Bakelite)	1590–1760	122	0.15
Nylon 6,6	1670	144	0.24
Polyisoprene	—	220	0.14

THERMAL EXPANSION

Thermal Expansion

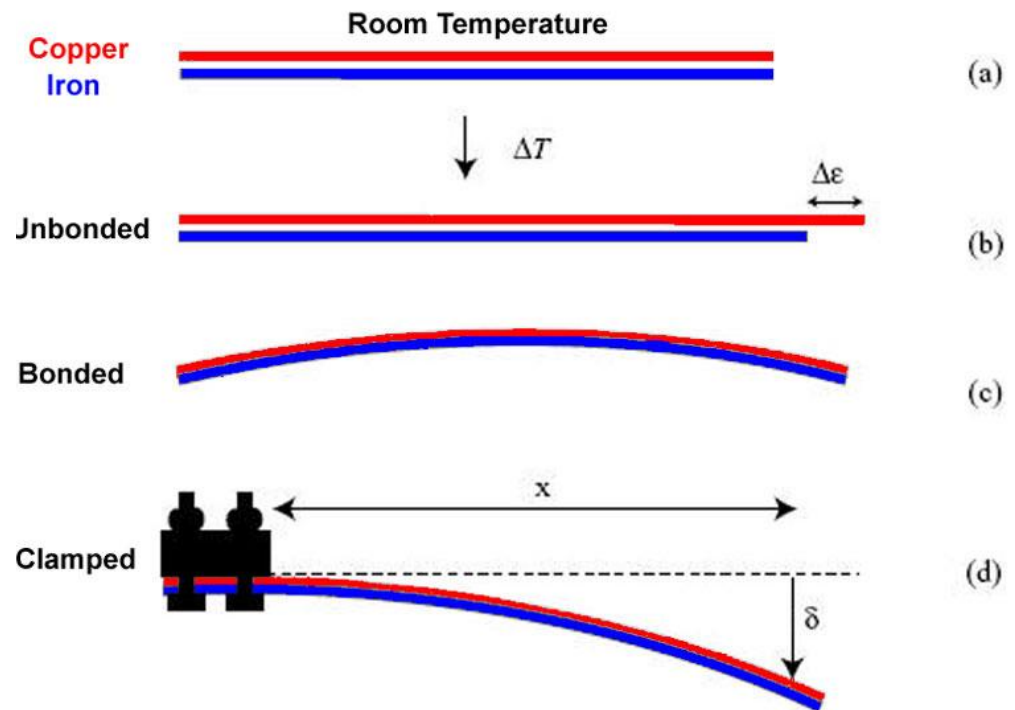
Bimetals: two metals bonded together with different coefficients of expansion

A **bimetallic strip** is used to convert a temperature change into mechanical displacement



Bimetal Strip

Two Metals Bonded Together with Different Coefficients of Expansion

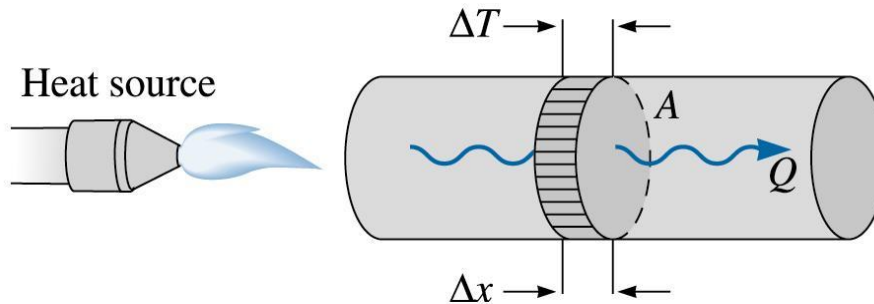


Thermostats, Thermometers, clocks, electrical devices

THERMAL CONDUCTIVITY

Thermal conductivity - A **microstructure-sensitive** property that measures the rate at which heat is transferred through a material.

When one end of a bar is heated, a heat flux Q/A flows toward the cold ends at a rate determined by the temperature gradient produced in the bar



$$q = \frac{Q}{A} = -k \frac{\Delta T}{\Delta x}$$

q = heat flux per unit area (W/m^2)

Q = heat flux (W)

K = thermal conductivity (W/m K)

ΔT = thermal gradient

Thermal energy is transferred by **two** mechanisms:

Transfer of free electrons

k_{e^-}

Lattice vibrations (phonons)

k_{ph}

$$k = k_{e^-} + k_{ph}$$

Conductive materials



transfer of free electrons $k \approx k_{e^-}$

Insulator materials



transfer of phonons $k \approx k_{ph}$

THERMAL CONDUCTIVITY

Relation between thermal and electrical conductivity :

$$L = \frac{k}{\sigma T}$$

L = Lorentz number (constant)

$$5.5 \times 10^{-9} \text{ (cal } \Omega \text{) / s K}^2$$

$$2.44 \times 10^{-8} \text{ (W } \Omega \text{) / K}^2$$

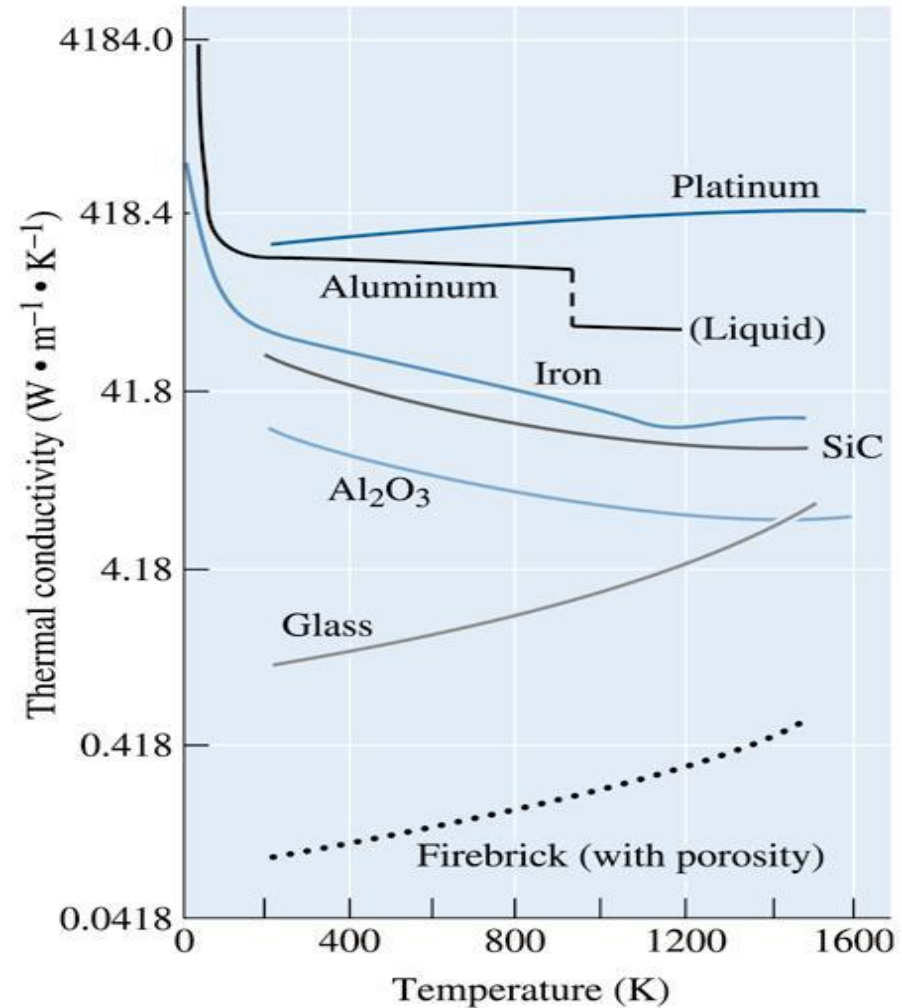
Material	Thermal Conductivity (W · m ⁻¹ · K ⁻¹)	Material	Thermal Conductivity (W · m ⁻¹ · K ⁻¹)
Pure Metals:		Ceramics:	
Ag	430	Al ₂ O ₃	16–40
Al	238	Carbon (diamond)	2000
Cu	400	Carbon (graphite)	335
Fe	79	Fireclay	0.26
Mg	100		
Ni	90	Silicon carbide	up to 270
Pb	35	AlN	up to 270
Si	150	Si ₃ N ₄	up to 150
		Soda-lime glass	0.96–1.7
Ti	22	Vitreous silica	1.4
W	171	Vycor™ glass	12.5
Zn	117	ZrO ₂	4.2
Zr	23		
Alloys:		Polymers:	
1020 steel	100	6,6-nylon	0.25
3003 aluminum alloy	280	Polyethylene	0.33
304 stainless steel	30	Polyimide	0.21
Cementite	50	Polystyrene	0.13
		Polystyrene foam	0.029
Cu-30% Ni	50	Teflon	0.25
Ferrite	75		
Gray iron	79.5		
Yellow brass	221		

Note: 1 cal/cm · s · K = 418.4 W · m⁻¹ · K⁻¹

THERMAL CONDUCTIVITY

The effect of **temperature** on the thermal conductivity (for metals) :

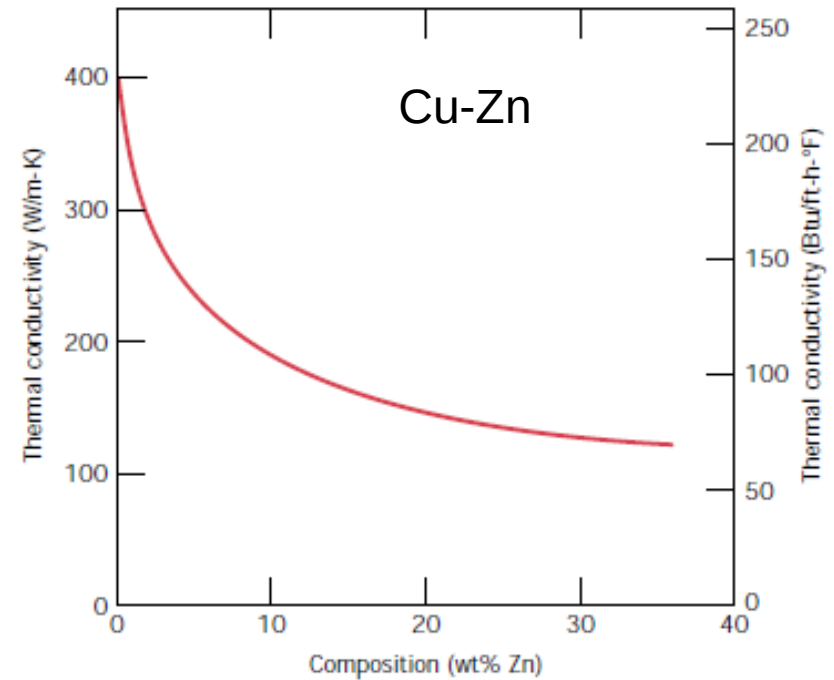
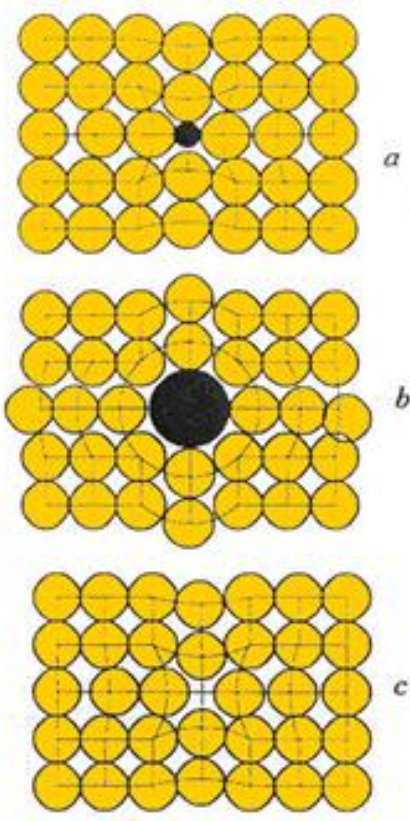
- ❑ Increasing temperature **increases** the contribution from lattice vibrations $k_{ph} \rightarrow$ thermal conductivity **increases**
- ❑ Increasing temperature **decreases** electron mobility (more scattering of electrons) $\rightarrow$ thermal conductivity **decreases**



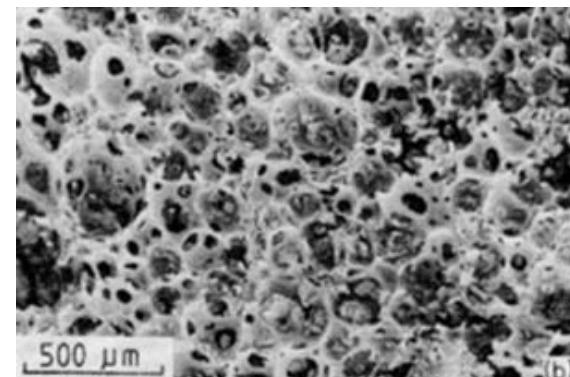
THERMAL CONDUCTIVITY

The effect of defects on the thermal conductivity

□ Alloying, defects, porosity: reduces thermal conductivity



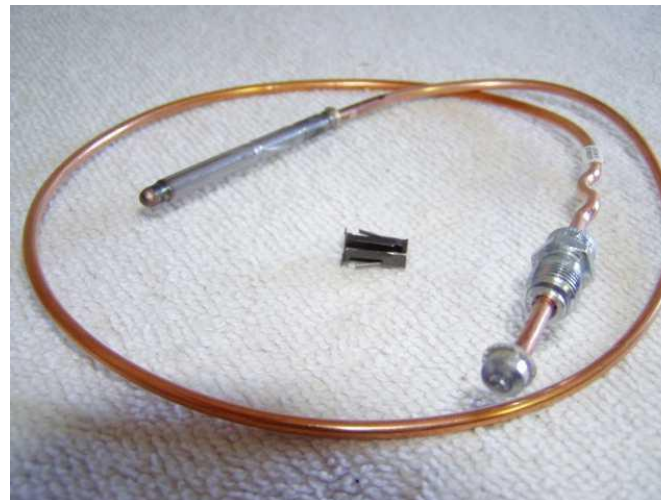
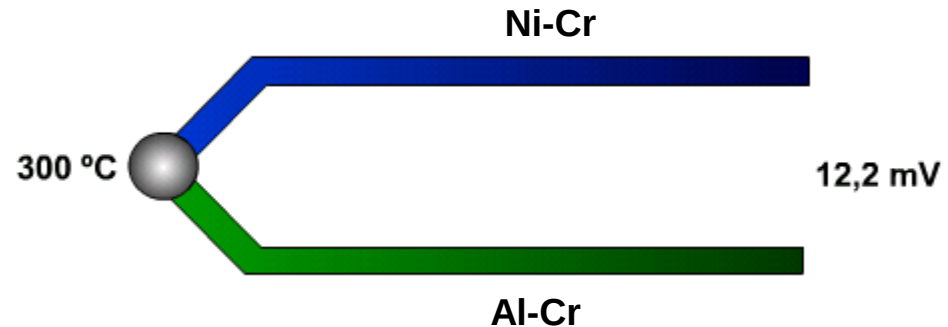
Porosity in ceramics



THERMAL CONDUCTIVITY

Termocouple

Temperature-measuring device consisting of two dissimilar conductors that contact each other at one or more spots. It produces a voltage when the temperature of one of the spots differs from the reference temperature at other parts of the circuit



THERMAL SHOCK

Thermal stresses - Stresses introduced into a material due to differences in the amount of expansion or contraction that occur because of a temperature change.

Confined materials

Stresses resulting from restrained thermal expansion or contraction

$$\sigma = E\alpha(T_0 - T_f) = E\alpha\Delta T$$

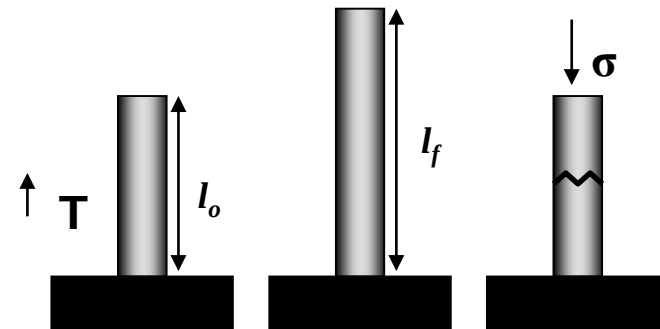
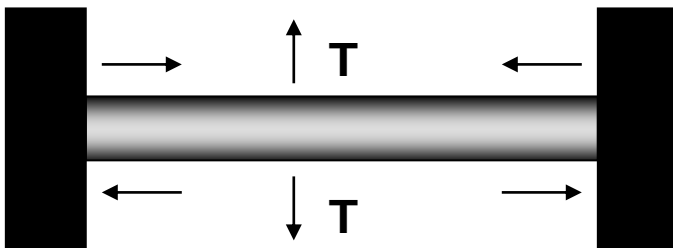
E = Young's Modulus (MPa)

α = linear coefficient of thermal expansion ($^{\circ}\text{C}^{-1}$)

σ = thermal stresses (MPa)

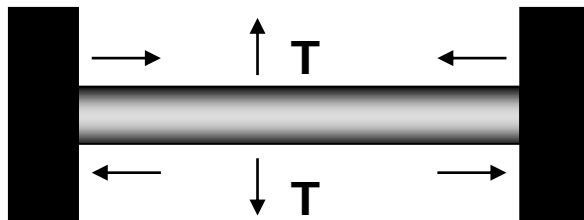
ΔT = thermal gradient

- Heating: $T_f > T_0 \rightarrow$ stress is **compressive** ($\sigma < 0$)
- Cooling: $T_f < T_0 \rightarrow$ stress is **tensile** ($\sigma > 0$)



Problem 5. Topic Mechanical properties

A steel wire, initially 1 meter in length, is stretched by a force of 1400 N at a temperature of 20 °C. After the force is applied, the wire is held at a constant length and then heated to 40 °C. Determine the required diameter of the wire to ensure that the thermal expansion stress compensates for the applied stress.



$$\sigma_{thermal} = E\alpha(T_0 - T_f) = E\alpha\Delta T$$

E = Young's Modulus (MPa)

α = linear coefficient of thermal expansion ($^{\circ}\text{C}^{-1}$)

σ = thermal stresses (MPa)

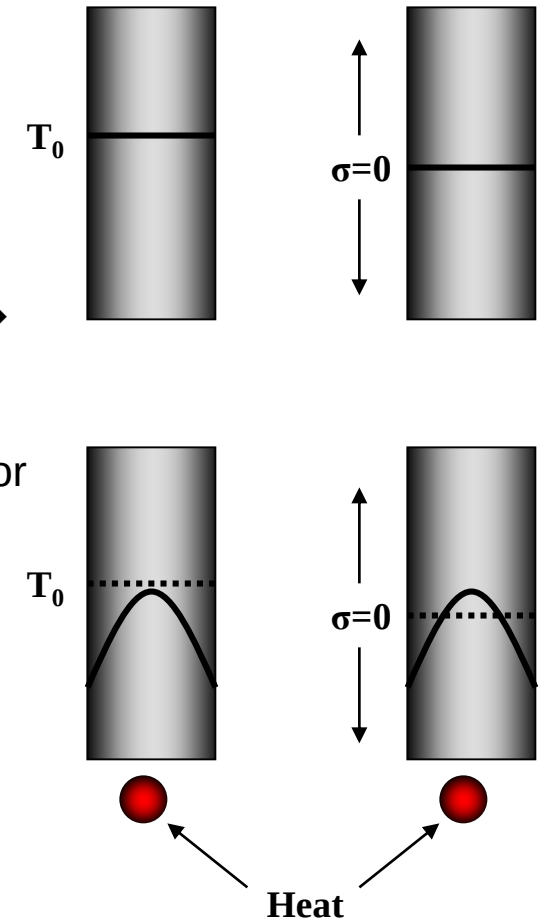
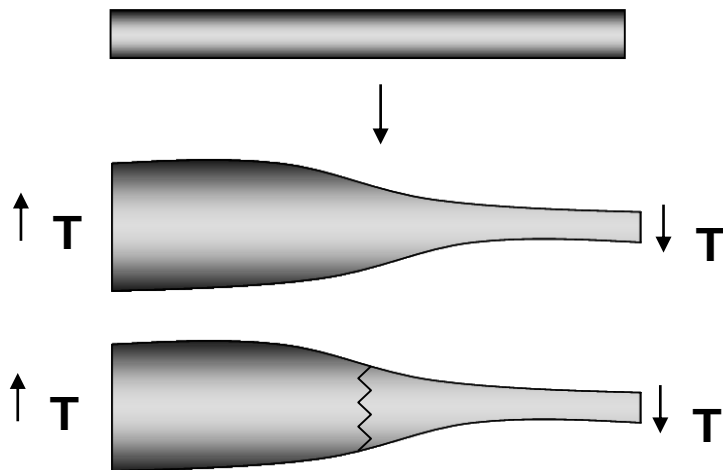
ΔT = thermal gradient

THERMAL SHOCK

Temperature gradients

Thermal stresses established as a result of temperature gradients across a body caused by rapid heating or cooling (the outside temperature changes more rapidly than the interior)

- Rapid heating: exterior of the specimen is hotter and expands → **compressive** surface stresses **tensile** interior stresses
- Rapid cooling: → **tensile** surface stresses **compressive** interior stresses



THERMAL SHOCK

Thermal Shock of brittle materials

Failure of a material caused by stresses introduced by sudden changes in temperature.

Thermal Shock is affected by several factors

$$TSR = \frac{\sigma_f \cdot k}{E \cdot \alpha_l}$$

Coefficient of thermal expansion: $\downarrow \alpha \Rightarrow \uparrow$ Thermal Shock resistance

Thermal conductivity $\uparrow k \Rightarrow \uparrow$ Thermal Shock resistance

Modulus of elasticity: $\downarrow E \Rightarrow \uparrow$ Thermal Shock resistance

Fracture stress: $\uparrow \sigma \Rightarrow \uparrow$ Thermal Shock resistance

Phase transformations: Dimensional changes $\Rightarrow \uparrow$ Thermal Shock